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From point sensing to intelligent systems: a comprehensive review on advanced sensor technologies for soil health monitoring

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Abstract

Soil health monitoring plays a critical role in enhancing crop productivity, mitigating adverse environmental impacts, and promoting sustainable agricultural practices. This review synthesizes recent advancements in sensor system technologies, examining their transformative potential for precision agriculture. It explores three interconnected themes: (i) the significant enhancement of sensor accuracy, specificity, and field-deployability for parameters such as moisture, nutrients, pH, and temperature through innovations in material science and transducer design; (ii) the capacity of integrated platforms, including Wireless Sensor Networks (WSN), the Internet of Things (IoT), and advanced spectroscopy, to overcome the spatial and temporal limitations of conventional point-based sensing; and (iii) the primary technological and scalability challenges that must be addressed to transition these technologies from research to widespread, practical adoption. Findings reveal that advancements in microelectromechanical systems (MEMS), nanotechnology, and polymer-based sensors have markedly improved sensitivity and cost-effectiveness, enabling rapid in-situ detection of NPK nutrients and reducing reliance on laboratory methods. Dielectric-based moisture sensors (TDR, FDR) and remote sensing techniques have proven highly effective for real-time irrigation management, while ion-sensitive field-effect transistors (ISFETs) and optical systems have enhanced nutrient profiling through improved pH and EC monitoring. The integration of these sensors into IoT-enabled networks facilitates extensive, real-time data collection and predictive analytics, supporting data-driven decision-making. Despite these advancements, key challenges persist, including the need for soil-specific calibration, high costs, and limited connectivity in remote areas. The evolution of soil sensor technology is progressing from basic devices to sophisticated systems capable of detecting specific ions and biological processes. Future prospects hinge on developing durable, cost-effective sensors with universal calibration models, leveraging machine learning for predictive analytics, and employing biodegradable materials to enhance sustainability. This review underscores the pivotal role of sensor technologies in achieving sustainable soil management and highlights the

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interdisciplinary innovations required to bridge the gap between prototyping and large-scale implementation in global agriculture.

Keywords Soil sensors, Precision agriculture, Nutrient sensors (npk), Soil health monitoring, Nanosensors, Biosensors

1 Introduction

The role of soil health is crucial in enhancing agricultural productivity, environmental sustainability, and global food security. The world's population is expected to reach 9.7 billion by 2050, which places an enormous pressure on the demand for sustainable and effective agriculture [202]. Soil is the primary support base for plant growth, and it affects water holding, nutrient supply, and microorganism population, all of which influence plant growth and yield [176]. Soil health monitoring using sophisticated sensing technologies is now crucial for precision farming, allowing farmers to maximize the use efficiency of farm resources, improve crop growth and yield and reduce environmental effects like nutrient loss through runoff and soil erosion [197]. Recent developments in soil sensor technologies over the past decade have revolutionized the measurement of soil properties, including moisture, pH, electrical conductivity (EC), nutrients, temperature, and biological activity, providing real-time, high-resolution data to support informed decisions [195].

From time-consuming, lab-based techniques, soil sensors have developed into advanced, field-deployable devices that are connected to wireless sensor networks (WSNs) and Internet of Things (IoT) platforms [152]. The shortcomings of conventional techniques, which were frequently time-consuming, expensive, and impractical for large-scale applications, have been addressed by these developments, which have increased the precision, affordability, and scalability of soil monitoring [215]. Since 2013, there have been notable advancements in the accuracy and usability of the sensors used for soil moisture determination, which include tensiometers, dielectric-based sensors (such as Time Domain Reflectometry (TDR) and Frequency Domain Reflectometry (FDR) and remote sensing methods [98, 195]. Tensiometers, for instance, are commonly employed because they are easy to use and have an accuracy of ± 3% in measuring soil water potential, even though they require regular maintenance [2]. With advancements like ion-selective field-effect transistors (ISFETs) and optical sensors providing improved durability and precision for in-situ measurements, pH and EC sensors have also grown more reliable [213, 214]. Nutrient sensors for the optimization of fertilizer application have progressed from sophisticated conventional methods such as Kjeldahl digestion to advanced electrochemical and optical systems that offer real-time detection of NPK [82]. MEMS integration, polymer-based sensors, and nanomaterials have improved sensitivity even at lower costs [97]. Microfluidic devices based on MEMS allow for quick nitrite detection, and polymer sensors thereby providing affordable NPK monitoring [51]. Temperature sensors are now combined with moisture sensors to provide complete soil thermal information, and biological sensors such as aptamer and enzyme-based biosensors track microbial processes and identify pathogens for integrated soil health evaluation [82, 206]. Emerging technologies such as WSNs, IoT integration, and hyperspectral imaging have revolutionized soil monitoring through extensive, real-time data collection [1]. IoT platforms enable remote access and predictive analytics via mobile devices, while WSNs facilitate continuous, in-field monitoring of soil parameters [155].

While drone and satellite hyperspectral photography produce high-resolution soil variability maps for precision agriculture, spectroscopy-based sensors employing NIR and MIR wavelengths provide quick, non-destructive soil analysis [142, 159].

Despite significant advancements, several challenges persist in the deployment of soil sensors, including the need for calibration across diverse soil types, high costs, and limited connectivity in remote areas [1]. Current review is focused on developing durable, cost-effective sensors with universal calibration models, integrating machine learning for predictive analytics, and employing biodegradable materials to enhance sustainability [92, 113]. The increasing importance of technology in meeting the demands of sustainable agriculture is reflected in the projected growth of the global precision agriculture market, which is expected to reach USD 34.9 billion by 2027. This review paper synthesizes the principles, applications, and limitations of recent innovations in soil sensor technologies. It highlights cutting-edge tools such as WSNs, IoT, and hyperspectral imaging examines sensors designed to monitor biological activity, nutrient levels, temperature, pH, electrical conductivity (EC), and soil moisture. This study posits that advanced sensor system technologies are fundamentally transforming soil health monitoring by enhancing the precision, scalability, and affordability of data acquisition, thereby enabling truly sustainable agricultural management and informed decision-making. In surveying the current scientific literature, we were particularly interested those pieces of work that are aligned with our research questions (RQs):

RQ1: How have advancements in material science and transducer design enhanced the accuracy, specificity, and field-deployability of modern soil sensors for key parameters like moisture, nutrients, pH, and temperature?—(Addressed in Sect. 2).

RQ2: To what extent can integrated platforms like Wireless Sensor Networks (WSN), the Internet of Things (IoT), and advanced spectroscopy overcome the spatial and temporal limitations of conventional point-based soil sensing?—(Addressed in Sect. 3).

RQ3: What are the primary technological and scalability challenges that must be addressed to transition advanced soil sensing technologies from research prototypes to robust, cost-effective, and widely adopted tools for sustainable soil management?—(Addressed in Sect. 4).

2 Types of soil sensors and their applications

To address our first research question (RQ1), we now present a critical examination of the fundamental types of soil sensors and their evolving applications. This section is structured to systematically evaluate the technological progression for each key soil parameter: moisture, pH/EC, nutrients, temperature, and biological activity. Our analysis will dissect how innovations, from the molecular level with novel polymers and nanomaterials to the systems level with solid-state and optical transducers, have collectively transformed sensing capabilities. We will explore the transition from laboratory-bound instruments to robust, field-deployable units, highlighting how these advancements have directly mitigated traditional limitations of accuracy, interference, calibration drift, and operational longevity. By reviewing each sensor category in detail, we aim to provide a comprehensive answer to RQ1, establishing a foundation for understanding the current state of the art in precision soil sensing.

2.1 Soil moisture sensors

Soil moisture is one of the most important soil parameters that define the degree of wetness or dryness, influencing the overall plant development. The importance of water has always been crucial in supporting various physiological processes of crops [187, 195]. The evolution of soil moisture monitoring technologies from initiation to till date has seen significant improvements in accuracy, affordability, and ease of deployment across various soil depths and conditions [138]. Generally, soil moisture sensors mainly composed of three parts: electronic circuits or devices for signal execution, communication circuits (such as coaxial cables), and probe electrodes embedded in the soil to provide bias and readout. Modern soil moisture detectors measure changes in soil properties that indicate water content, such as dielectric permittivity, matrix possibilities, and mass (volumetric moisture level) of soils, to determine the water level in soil [98]. Soil moisture sensors can be divided into four main subcategories based on their working principles: remote sensing techniques, laboratory-based techniques, on-site techniques, and in-field techniques which are further elaborated [54, 169].

At-site soil moisture monitoring methods can provide soil moisture measurements at different depths, are more easily calibrated, have time scale control and generally easy to use for farmers. They have the potential to provide real-time point-based high-resolution moisture content data that is representative of plot-to-field-scale areas, especially helpful for agricultural applications. To illustrate the practical application of these methods, the following instruments are commonly used in field-based soil moisture monitoring.

2.1.1 Tensiometer/irrometer

A tensiometer is a widely used instrument for measuring soil water tension, particularly in irrigated agriculture. Structurally, it consists of a pressure gauge mounted at the top, a porous ceramic cup at the bottom, and a sealed stem filled with distilled water. When inserted into the soil, the porous cup allows water to move between the tensiometer and the surrounding soil, thereby equilibrating with the soil's moisture potential. The pressure gauge then reflects the tension or energy required by plant roots to extract water from the soil. Tensiometers are effective within a range of 0 to − 85 centibars and offer an accuracy of approximately ± 3% of the full scale. Tensiometers, along with granular matrix sensors, are among the most commonly adopted soil moisture sensors in commercial agricultural applications. Comparative studies have shown that soil moisture readings from tensiometers and soil moisture meters were 8.51% and 5.45% higher, respectively, than those obtained using the gravimetric method [112]. Further research by Shekhar et al. [168] demonstrated that soil moisture distribution patterns vary with emitter discharge rates. Specifically, horizontal water spread increased by 23.3% and 43.3% when discharge rates rose from 2 LPH to 4.4 LPH and 6.0 LPH, respectively, while vertical water movement decreased under higher discharge conditions. However, they come with several operational limitations. Regular maintenance is required, including refilling or replacing the distilled water every two to four weeks, depending on soil type and irrigation frequency. Additionally, the device must be removed during winter to prevent damage from freezing temperatures. Careful installation is essential, as the presence of air pockets around the ceramic cup can significantly reduce measurement accuracy [98]. Figure 1a shows the Irrometer and Fig. 1b shows the Watermark.

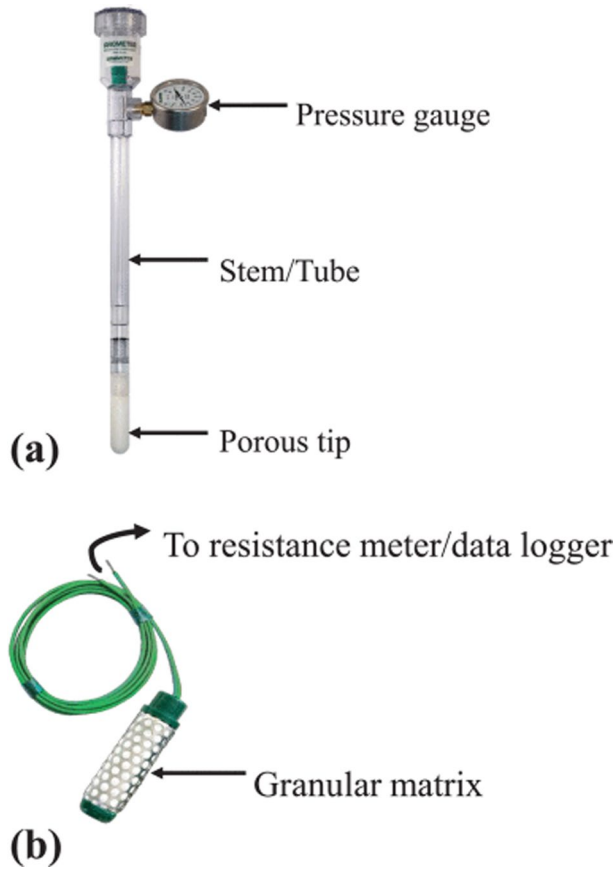

Fig. 1 **a** Irrometer—a commercial tensiometer. **b** Watermark—a granular matrix SWP sensor. (The Irrometer Co., Riverside, CA; http://www.irrometer.com)

2.1.2 Gypsum block or granular matrix sensor

Gypsum block and granular matrix sensors are cost-effective tools for monitoring soil moisture, particularly in the root zone of crops. These sensors utilize a porous medium typically a block of gypsum ($CaSO_4{\cdot}2H_2O$) or a granular matrix embedded with electrodes. The sensor is buried in the soil, where it functions by detecting changes in electrical resistance as water infiltrates the porous material. The working principle is based on the solubility of gypsum in water. As soil moisture percolates through the porous matrix, the gypsum gradually dissolves, releasing ions that reduce the electrical resistance between the embedded electrodes. This change in resistance is then correlated with soil moisture levels [91]. The dissolution of gypsum plays a buffering role, stabilizing the sensor's response to moisture fluctuations [52]. However, over time, degradation of the gypsum block may affect long-term accuracy and necessitate replacement. Despite this limitation, the primary advantage of these sensors lies in their affordability, ease of fabrication, and suitability for repeated measurements at fixed locations. Ezekiel et al. [58] emphasized their widespread use in field conditions due to their simplicity and low operational cost, making them particularly valuable for smallholder and resource-constrained agricultural systems.

2.1.3 Thermal probe/heat pulse method

Thermal soil moisture sensing probes operate on the principle of heat dissipation through soil or a porous medium, which is directly influenced by the moisture content. These sensors typically monitor the temperature of either the soil itself or a porous block (commonly made of gypsum or ceramic) before and after the application of a brief heat pulse. The rate of heat transfer is governed by the thermal conductivity of the medium, which increases with higher soil moisture levels. The core components of these sensors include a heating element often a thermistor and a temperature sensor, such as a pn junction. Depending on the design, these elements may be embedded directly in the soil to measure Soil Moisture Content (SMC) or housed within a porous block to estimate Soil Water Potential (SWP). It can be of different types like single, dual or multi-probe designs. A single-probe heat pulse (SPHP) is based on a single npn bipolar junction that functions as the heating element and the base-emitter junction as the temperature sensor. Dual probe heat pulse (DPHP) consists of a separate heater and temperature monitoring elements [153]. In contrast, a multi-probe heat pulse sensor is surrounded by several temperature sensors [66]. Commercially available thermal soil moisture sensors offer a relatively low-cost solution with an accuracy range of 5–10% [163]. However, their performance is subject to several limitations. These include the need for soil-specific calibration, slow response times, sensitivity to ambient temperature and humidity fluctuations, and high energy consumption during operation.

2.1.4 Neutron probe method

The neutron scattering method, first elucidated by Gardner and Kirkham [70], is a non-destructive technique for measuring soil moisture content. It operates by emitting fast neutrons from radioactive sources such as radium-beryllium or americium-beryllium. These neutrons collide with hydrogen nuclei primarily found in soil water and are subsequently slowed (thermalized). The density of thermalized neutrons detected by the probe correlates with the volumetric water content of the soil. When properly installed by fully burying the probe with minimal air gaps and calibrated for soil-specific parameters such as bulk density and metal concentration, neutron probes offer continuous and accurate monitoring with a precision range of 1–5% [39]. They are largely unaffected by temperature and salinity, making them reliable under varying environmental conditions. However, limitations include poor vertical resolution, difficulty in measuring surface layers, and sensitivity to soil texture and density variations. Despite their high accuracy (± 0.02 in volumetric water content under optimal calibration) [50], the widespread adoption of neutron probes is constrained by high costs, radiation safety concerns, and the need for trained personnel. The reliance on radioactive materials further complicates field deployment and regulatory compliance [88]. Neutron scattering remains a valuable tool in research and high-precision agricultural applications, particularly where deep soil moisture profiling is required. Figure 2 shows the schematic of a neutron probe SMC sensor.

2.1.5 Dielectric methods (TDR, FDR & SWR)

The dielectric constant of soil is influenced by several factors, including temperature, salinity, bulk density, volumetric water content, electromagnetic frequency, particle geometry, and moisture distribution. Among these, soil water content plays the most

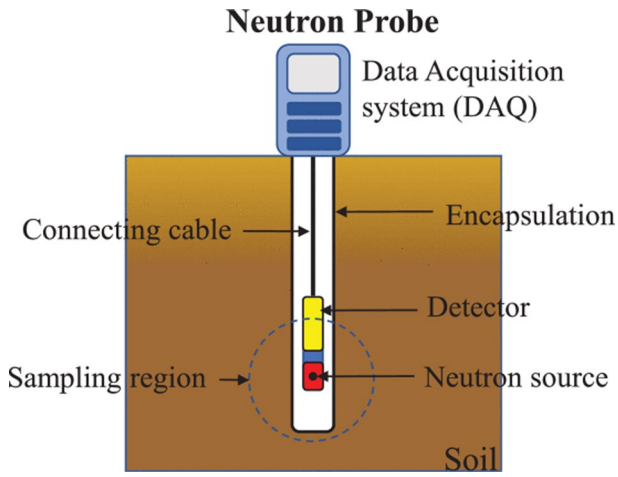

Fig. 2 Schematic of a neutron probe SMC sensor deployed in the field

significant role due to the markedly higher dielectric constant of water compared to air and other soil constituents. Consequently, measuring the dielectric constant provides an effective means of estimating soil moisture levels. Dielectric-based sensing techniques are widely adopted in precision agriculture for their non-destructive nature and real-time capabilities. The most commonly employed methods include Time Domain Reflectometry (TDR), Frequency Domain Reflectometry (FDR), and Standing Wave Ratio (SWR), each offering distinct advantages in terms of sensitivity, depth profiling, and ease of integration [114].

TDR is a widely recognized technique for high-accuracy soil moisture measurement, based on the principle that the propagation velocity of electromagnetic waves varies with the dielectric properties of the medium [185]. In TDR systems, a pulse signal is transmitted from a signal generator through a coaxial cable to a probe inserted into the soil. At the interface between the cable and the probe, an impedance mismatch occurs, causing part of the signal to reflect while the remainder continues along the probe. The travel time of the reflected signal is used to calculate the soil's apparent dielectric constant, which correlates strongly with volumetric water content [195]. Compared to capacitive sensors, TDR sensors are more expensive but offer superior accuracy and are less susceptible to interference from temperature fluctuations and soil salinity [83, 84]. Recent advancements have introduced multiplexing capabilities, allowing a single TDR unit to simultaneously monitor soil moisture across multiple depths, thereby enhancing spatial resolution and data efficiency [195]. Introduction of probe technology to TDR devices has led to the construction of multi-depth, down-hole sensors like the Campbell Scientific SoilVUE 10. They can measure soil moisture along with properties like permittivity, EC, and temperature, at varying depths. With just one sensor unit, soil health can be monitored easily from the surface through the root zone. We can actually monitor how soil is distributed at different depths, which makes irrigation, and water management more optimised. For example, if there is a dry layer present in depth, but at the surface, the soil seems moist, it will send a signal that roots are not getting enough water. Such a crucial role makes these sensors useful for irrigation scheduling and research works, enabling precise tracking of soil water [80].

FDR offers greater flexibility than TDR in terms of operating frequency and probe geometry, making it well-suited for diverse soil conditions and sensor configurations

[208]. The principle underlying FDR is based on the relationship between soil moisture and the sensor's equivalent capacitance: as moisture content increases, the dielectric constant of the soil rises, thereby increasing the capacitance and altering the sensor's resonant frequency. The sensor operates by transmitting high-frequency signals typically in the range of tens to hundreds of megahertz via a high-frequency oscillator. Changes in the dielectric properties of the soil affect the resonance frequency, which is detected by a frequency-sensitive circuit and used to infer volumetric water content [217, 218]. The capacitance measured between the probe electrodes is directly proportional to the dielectric constant of the surrounding medium, enabling continuous monitoring of moisture dynamics. Recent advancements in FDR technology include improved temperature compensation mechanisms and enhanced calibration algorithms, which contribute to greater measurement stability and accuracy under variable environmental conditions [206]. With automated calibration containing smart algorithms and microcontrollers, these sensors are flexible to all soil textures, enhancing accuracy in varied field conditions [80]. Multimodal technology has made FDR devices able to detect pH, temperature, and EC together with soil moisture content, leading to well-planned irrigation and precise soil health management [115, 215]. Data collection and feedback can be streamlined by sensors using wireless technologies like LoRa and Bluetooth to provide real-time readings from distant or large fields straight to automated irrigation systems [80, 83, 84]. These sensors are now available for both ordinary farm management and scientific study thanks to advancements in electronics that have reduced their size to inexpensive, small models, including those that work with Arduino [150, 215]. Additionally, these sensors are available in a range of sizes and forms to accommodate different soil profiles and crop varieties, and novel calibration methods handle issues caused by air gaps or soils with high salinity or organic content [9, 80]. All things considered, these advancements have significantly enhanced the function of FDR sensors in contemporary precision farming.

The SWR of a transmission line can be greatly impacted by variations in the dielectric constant of a three-state combination. Similar to TDR but measuring SWR rather than time difference (ΔT), Yu et al. [209] suggested a method for assessing soil moisture based on SWR in microwave theory. Although this sensor is less accurate and replaceable, it is less expensive and simpler to install. A standing wave is created when a 100 MHz signal is transmitted to a probe via a coaxial wire and reflected by an impedance mismatch. The dielectric constant of the soil affects the voltage differential, which changes with probe impedance [182, 183]. Recent advancements in SWR sensors include advanced temperature-compensated models that are known for boosted accuracy and a rapid dynamic response, often achieving readings in less than one second. Field trials done in China and Germany confirm that these sensors deliver constant measurements across different soil textures, signifying their international equivalence and dependability. Besides, the integration of AI and supply chain management systems has enhanced their adoption, with new SWR sensor models increasingly featuring built-in, software-driven self-calibration to ensure easy handling and reliable performance [216].

2.1.6 Capacitance-based sensors

Capacitive sensors measure soil moisture by detecting changes in capacitance, eliminating the need for exposed metal parts, which makes them durable and widely used in the

field. [61]. Capacitive sensors have gained popularity due to their relatively low cost, ease of installation, and reasonable accuracy for most agricultural applications [82]. According to Abdulraheem et al. [4], the dielectric constant in the soil varies proportionately with its moisture content, which is why the capacitive sensors showed an almost proportional interaction between the measured capacitance and the soil moisture level. According to Suleiman et al. (2009), a proposed moisture detector is a fringing electric field-based device that can detect moisture within a range of 1–80% of the volumetric water level and shows linear responses. The sensor's measurement of capacitance was instantly connected to the input of the capacitance-using device into a digital conversion device. Due to the small size of the sensor and its imprecise measurements caused by variations in electrode thickness and geometry, the main drawback of the sensor appears to be its low infiltration depth [178].

2.2 Remote sensing-based sensors

Remote sensing (RS) techniques utilize reflected or emitted electromagnetic (EM) energy for measuring soil moisture content. These techniques are separated into two categories: (i) Active, which uses radiometers to measure reflected or scattered energy, and (ii) Passive, which uses radiometers to detect emitted radiation (brightness temperature). Remote sensing-based soil moisture content (RS-SMC) retrieval encompasses a range of techniques across different segments of the electromagnetic spectrum. These methods vary in their sensitivity to surface and subsurface moisture, spatial resolution, and susceptibility to environmental factors such as vegetation cover, surface roughness, and atmospheric interference. Depending on the spectral domain and sensing mechanism, RS approaches are broadly categorized into optical, thermal, and microwave methods each offering distinct advantages for specific soil and climatic conditions.

2.2.1 Optical methods (visible/near-infrared)

This technique uses near-infrared (IR) and visible (0.4–2.5 μm) reflectance spectroscopy, where water molecules show clear absorption bands (e.g., at 1.94 μm). Skin SMC has a linear relationship with reflectance amplitude. High spatial resolution is offered by this early RS-SMC technology; however, it is only applicable to surface (skin) measurements. Only superficial SMC measurements may be made with this technique. Reflectance as a factor affecting the soil moisture has been recognised by many scientists. A physical method based on the analysis of four different type of soil at varying moisture content was conducted to establish the relation of soil reflectance variations due to moisture change, as given by [166]. [122] designed an index that was able to sense both soil and vegetation water content and called it the normalized multi band drought index (NMDI).

2.2.2 Thermal-infrared (IR) methods

It uses thermal properties of soil in the 3.5–14 μm range. The two main techniques used in thermal infrared remote sensing for determining soil moisture are based on measurements of soil surface temperature and operate in the 3.5–14 μm wavelength range. [13] Because higher moisture content areas remain cooler during the day and warmer at night due to increased thermal inertia (resistance to temperature change), the thermal inertia method takes advantage of the relationship between soil moisture and diurnal temperature variations. This simple method is very accurate, but it is only applicable to

areas with little vegetation cover. With applications combining MODIS data to achieve daily 1 km resolution soil moisture products, the vegetation index method makes use of the intricate relationship between soil moisture, normalized difference vegetation index (NDVI), and land surface temperature (LST), which researchers refer to as the "Universal Triangle" relationship [193]. Even though both strategies work well and have obvious physical benefits, they have similar drawbacks, such as shallow soil penetration, cloud contamination, and reliance on regional weather factors like humidity, wind speed, and air temperature, which affect accuracy over time and across different types of land cover.

2.2.3 Microwave methods

Microwave sensing techniques, operating within the 1 mm–30 cm wavelength range, enable deep penetration through soil and vegetation and are unaffected by cloud cover [106]. These methods are broadly classified into active and passive systems. Active microwave sensors (e.g., radars) emit microwave signals and measure the backscatter from the ground surface. While offering high spatial resolution, their performance is sensitive to surface characteristics such as soil moisture, vegetation type, and surface roughness. Backscatter modeling commonly employs the integral equation model, alongside empirical, semi-empirical, and machine learning-based approaches [196].

Passive microwave sensors (e.g., radiometers) detect naturally emitted radiation from the soil, quantified as brightness temperature (TB). TB is proportional to the product of surface temperature (Tsurf) and emissivity (e) [121]. Modeling typically involves: (i) a radiative transfer model, linking TB to the soil dielectric constant (εs), and (ii) a dielectric mixing model, relating volumetric water content (VWC) to εs. While active sensors are more affected by surface roughness, passive sensors exhibit greater stability under such conditions. Integrating data from both sensor types enhances soil moisture retrieval by leveraging complementary land surface microwave signatures [7].

2.3 Proximal In-field soil moisture monitoring

Proximal in-field soil moisture sensors estimate soil moisture content (SMC) without direct contact, operating near the soil surface to capture spatial variability at plot and field scales [20]. These systems include both terrestrial and aerial platforms, with unmanned aerial vehicles (UAVs) increasingly employed for rapid, non-invasive assessment. To enhance spatially distributed, non-contact soil moisture estimation, various proximal sensing technologies have been developed, each leveraging distinct physical principles and operational platforms.

2.3.1 Ground penetrating radar (GPR) system

A drone-mounted soil moisture sensing system integrates a vector network analyzer (VNA) with a hybrid horn-dipole antenna operating in the 500–700 MHz frequency range. This setup estimates soil moisture at depths of 10–20 cm by measuring reflection coefficients from the subsurface. The data processing involves two key stages: (i) full-wave inversion using weighted least-squares optimization to retrieve soil dielectric properties, and (ii) radar modeling via transfer functions to compute the Green's function of the air–subsurface system. The derived relative dielectric permittivity is then converted to volumetric water content using established empirical relationships. This technology has successfully generated high-resolution soil moisture maps across three

Belgian agricultural fields, demonstrating its potential for non-invasive, spatially extensive moisture profiling.

2.3.2 Multi-spectral and thermal imaging systems

Multi-spectral and thermal imaging systems, particularly when deployed via unmanned aerial vehicles (UAVs), have become essential tools for high-resolution soil moisture and crop stress assessment in precision agriculture. UAVs offer superior spatial resolution compared to satellite platforms due to their low-altitude operation, enabling detailed monitoring at the plot and field scale. Multi-spectral sensors capture reflectance across visible and near-infrared bands, allowing the calculation of vegetation indices such as NDVI and SAVI, which are used to develop agricultural water stress models. Studies have reported correlation coefficients (R^2) ranging from 0.47 to 0.81 depending on the index and crop growth stage [114]. Thermal imaging complements this by detecting canopy temperature variations, which are indicative of water stress. Additionally, machine learning approaches such as artificial neural networks (ANNs), incorporating inputs like vegetation indices, thermal data, and spectral bands, have demonstrated improved predictive accuracy ($R^2 \approx 0.77$) and robustness compared to traditional index-based methods [206]. These systems collectively enhance decision-making in irrigation scheduling and stress detection.

2.3.3 EMI-based moisture sensors

Electromagnetic induction (EMI)-based methods typically operate at lower frequencies, such as the 38 kHz used in the EM38 system [57]. This technique avoids the logistical challenges of down-tube deployments and core drilling by employing horizontally spaced coils to measure soil conductivity. EMI systems offer several advantages, including rapid data acquisition, high spatial precision, and non-invasive access to deeper soil layers. Despite these strengths, their application in intelligent agriculture remains limited due to high equipment costs and susceptibility to electromagnetic interference. In contrast, soil ultrasonic sensors estimate moisture content by analyzing the acoustic disparity between forward and backward ultrasonic wave propagation. These sensors require both an ultrasonic source and a detector, which slightly increases their cost. The time differential between reciprocating signals reflects fluctuations in soil water content. Additionally, the dielectric constant or observable permittivity of soil is directly correlated with its moisture level, providing a reliable metric for assessment [177].

2.3.4 Ultrasonic-based moisture sensors

Ultrasonic soil moisture sensors typically consist of a brass or stainless-steel waveguide pipe, with an ultrasonic transmission device securely mounted at the upper end. These sensors generate ultrasonic waves commonly at 40 kHz via a transducer and utilize reflection techniques to assess soil moisture by analyzing wave scattering at the soil surface, which varies with moisture content [198]. When installed underground, ultrasonic sensors enable efficient multipoint monitoring across large agricultural landscapes. However, temperature fluctuations significantly affect measurement accuracy, necessitating the implementation of artificial pair systems to compensate for thermal variability. Despite their technical advantages, ultrasonic sensors such as the UGT582 (priced at approximately $173) remain considerably more expensive than conventional conductive

soil moisture sensors, like the Adafruit STEMMA detector, which costs around $7.50. This cost disparity continues to hinder the widespread adoption of ultrasonic sensing technologies in advanced agricultural systems [60].

2.4 Soil pH and EC sensors

Soil pH sensing technologies have evolved significantly, offering diverse approaches tailored to precision agriculture, environmental monitoring, and industrial applications. These sensors are broadly classified into three categories: conventional glass electrodes, optical sensors (including fluorometric and ratiometric systems), and electric sensors such as ISFETs, potentiometric, and conductometric devices. Advanced sensing platforms have further expanded this landscape, incorporating infrared spectroscopy, acoustic cantilever-based sensors, remote sensing, and wireless networks for large-scale deployment [53]. Infrared spectroscopy enables non-destructive pH measurement by analyzing the reflectance of absorbed infrared light from soil surfaces, suitable for both laboratory and field conditions [127]. Remote sensing techniques utilize satellite imagery to estimate soil pH based on surface reflectance characteristics, facilitating regional-scale assessments [36]. Acoustic pH sensors, particularly cantilever and microcantilever designs, operate on piezoresistive principles, where surface stress alters electrical resistance. These sensors are often fabricated using layered structures of silicon dioxide (SiO_2) and silicon nitride (Si_3N_4), with functional modifications such as aminosilane-coated SiO_2/Au and Si_3N_4/Au to enhance sensitivity [26, 188]. Miniaturized pH sensors have gained traction in biomedical, biotechnological, and food processing sectors, especially in microenvironmental monitoring [33]. Hydrogel-based sensors, leveraging piezoresistive materials like polyvinyl alcohol-polyacrylic acid, demonstrate promising mechanical–electrical transduction capabilities [100]. Photolithographic techniques allow polymer-coated electrodes to be patterned on oxidized silicon, enabling scalable commercial production. A recent study tested a novel sensor for continuous soil pH monitoring across high-pH sandy loam and low-pH clay/volcanic sand soils. After a stabilization period of 12–24 h, the sensor delivered consistent readings, with coefficients of variation at 1.885 and 0.945%, and error rates of 7.37 and 6.02% compared to standard glass electrode benchmarks [55]. These findings underscore the potential of next-generation pH sensors to deliver reliable, real-time data for precision soil management.

2.4.1 Ion-selective field-effect transistors (ISFETs)

ISFETs represent a widely adopted class of ion-selective electrodes (ISEs), particularly for pH measurement. These potentiometric sensors incorporate selective membranes that minimize matrix interferences, making them suitable for complex soil environments. As solid-state devices, ISFETs offer high durability and long-term stability under harsh field conditions [83, 84]. Recent advancements have focused on improving gate materials and temperature compensation algorithms to enhance measurement accuracy and operational reliability [181]. The ability to perform in situ pH measurements without the need for labour-intensive sampling and laboratory analysis has made ISEs increasingly valuable for soil mapping in precision agriculture [190]. Current research continues to prioritize improvements in electrode durability and response time to support real-time, field-deployable sensing systems.

2.4.2 Optical pH sensors

Optical sensors assess pH variations using colorimetric or fluorescence indicators [206]. One benefit of these sensors is their ability to give wireless measurements via optical fiber connections and their immunity to electromagnetic interference [195]. Recent developments in optical methods include optical pH sensors (also called planar pH optodes), which consist of a sensitive dye (like methylene blue) enclosed in a matrix where protons exchange with each other, keeping the dye intact [173]. pH optodes are quite comparable to glass electrodes in design, and the range for pH optodes varies between 3 to 9 not needing a reference electrode. Some optical pH sensors utilise synthesized bases, such as 1,2-cyclohexanediamine, that are also found to respond optically relative to absorption and emission [10] with pH range varying between 3 and 7.8.

2.4.3 EC sensors

EC Sensors play a vital role in assessing soil salinity by measuring the ability of soil to conduct electrical current. Contemporary EC sensors utilize advanced electrode materials and signal processing techniques to improve accuracy and reliability. To minimize polarization effects and enhance measurement stability, four-electrode configurations have become standard practice [181]. Traditionally, EC testing has been employed to characterize soil properties, though earlier studies often overlooked the influence of measurement frequency. Recent advancements have introduced frequency selective EC techniques, which apply specific frequencies to improve the sensor's responsiveness to salinity variations. While EC measurement is inherently non-ion-selective, frequency tuning has enhanced its ability to differentiate salinity levels more precisely, making it increasingly valuable for precision agriculture applications.

2.5 Nutrient sensors

The use of fertilizers and nutrients is one of the most expensive aspects of crop production, both in terms of immediate cash expenditures and long-term environmental impacts [144]. Therefore, precise soil nutrient monitoring enables enhanced application efficiency and greater sustainability in agriculture. Conventional methods of soil testing involve sample extraction, preliminary treatment, and chemical analysis, which are time-consuming, costly, and require skilled operators. Techniques such as Kjeldahl digestion and Dumas combustion are commonly used for nitrogen detection but are limited by procedural complexity and incomplete nutrient profiling. To overcome these limitations, modern sensing methods have been developed, offering rapid, in-situ nutrient analysis with minimal sample preparation. These include optical, electrochemical, and conductivity-based approaches. Optical methods utilize reflectance spectroscopy, colorimetry, Near-Infrared Reflectance Spectroscopy (NIRS), and Laser-Induced Breakdown Spectroscopy (LIBS) for non-destructive, multi-element detection. Electrochemical methods employ ion-selective electrodes (ISEs) or ion-selective membranes (ISMs) for real-time nutrient quantification. Conductivity-based techniques estimate NPK levels by measuring ionic conductivity using electrodes and alternating current (AC) voltage [72]. Comparative analysis of contemporary in-situ point-based soil moisture sensing technologies are described in Table 1.

In addition to these core techniques, advanced technologies have further expanded the scope and precision of soil nutrient analysis. MEMS-based sensors offer high

Table 1 Comparative analysis of contemporary in-situ point-based soil moisture sensing technologies

Sensor types	Measurement principle	Key strengths	Key limitations and challenges	References
Tensiometer	Measures soil water potential (matric potential) via pressure difference across a porous ceramic cup in hydraulic equilibrium with the soil	Direct measurement of plant-available water; largely independent of soil texture, salinity, and temperature; provides high-quality data on water stress	Limited measurement range (0 to ~ − 85 kPa); requires frequent maintenance (de-airing, refilling); fragile; not suitable in freezing conditions; limited depth capability	[2, 112]
Gypsum block/granular matrix sensor	Infers soil water potential from changes in electrical resistance across a porous gypsum or granular matrix as it absorbs water	Low cost; simple design and operation; suitable for repeated measurements at fixed points; minimal maintenance	Slow response time; accuracy compromised by salinity and temperature fluctuations; gypsum dissolves over time, requiring replacement; requires soil-specific calibration	[48, 91]
Single-probe heat pulse (SPHP)	Estimates volumetric water content from the thermal response (heat dissipation) to a short heat pulse applied by a single probe containing both heater and thermometer	Low cost; small sensing volume; insensitive to soil salinity and texture; suitable for automated systems	High energy consumption per measurement; requires complex calibration accounting for soil specific heat capacity and thermal conductivity; sensitive to ambient temperature drift	[153]
Dual-probe heat pulse (DPHP)	Measures volumetric water content by analyzing the temperature rise over time at a specific distance from a linear heat source	Insensitive to soil salinity and composition; can simultaneously estimate soil thermal properties; open-source designs available	Energy intensive; slower response time limits temporal resolution; moderate accuracy; requires calibration for soil type	[66]
Neutron probe	Measures volumetric water content by detecting the density of thermalized neutrons after fast neutrons are scattered by hydrogen atoms (primarily in water)	Highly accurate; large measurement volume reduces small-scale variability; deep soil profiling capability; unaffected by salinity and temperature	High capital cost; radiation safety hazards require licensed operators; poor vertical resolution; cannot be left unattended; and regulatory restrictions	[39, 50]
Time domain reflectometry (TDR)	Determines volumetric water content by measuring the propagation velocity of an electromagnetic wave along a buried probe, which depends on the soil's dielectric permittivity	High accuracy; measures water content and bulk electrical conductivity simultaneously; minimal calibration needed for many soils; robust measurement principle	High cost per sensor point; complex electronics; installation can disturb soil; accuracy affected by high clay or organic matter content	[83, 84, 185]

Table 1 (continued)

Sensor types	Measurement principle	Key strengths	Key limitations and challenges	References
Frequency domain reflectometry (FDR)/capacitance	Estimates volumetric water content by measuring the operating frequency or capacitance of a circuit, which changes with the soil's dielectric permittivity	Moderate cost; capable of continuous, long-term logging; simple installation; suitable for multi-depth profiling	Requires soil-specific calibration; accuracy susceptible to soil salinity, temperature, and bulk density; sensitive to air gaps and soil-probe contact	[37, 215]
Standing wave ratio (SWR)	Infers water content from the standing wave ratio on a transmission line, which is affected by the soil's dielectric properties	Lower cost and simpler electronics than TDR; potential for low-power operation	Generally lower accuracy and stability compared to TDR/FDR; less common; requires extensive calibration	[71, 146]
High-frequency capacitance (e.g., Decagon EC-5)	A specific subset of FDR sensors operating at high frequency (~ 70 MHz) to minimize the influence of soil salinity and texture on the capacitance measurement	Excellent stability; fast response time; low power consumption; well-established performance for agricultural applications	Requires soil-specific calibration for highest accuracy; initial cost can be high with dedicated data loggers	[129, 130, 215]
Graphene-oxide based sensors	Measures changes in electrical conductivity or capacitance of a graphene-oxide film whose dielectric properties are highly sensitive to water molecule adsorption	Potential for very high sensitivity; rapid response; can be fabricated for low-cost and miniaturized form factors	Predominantly in research phase; long-term stability and drift in variable field conditions not yet fully validated; susceptibility to fouling	[97]
Acoustic/ultrasonic sensors	Infers water content by measuring the attenuation or velocity of sound waves propagating through the soil, which changes with moisture	Non-electromagnetic principle avoids dielectric-based limitations; potential for large sensing volume	Highly sensitive to soil texture, density, and contact; significantly affected by temperature; currently high cost and limited commercial availability	[60, 85]

surface-to-volume ratios and are used for rapid detection in microfluidic formats, such as nitrite sensing. Polymer-based sensors, utilizing materials like polypyrrole and polyurethane, provide cost-effective solutions for detecting nitrogen, phosphorus, potassium, and other parameters. Nanobiosensors operate at the atomic scale and offer ultra-sensitive detection, with potential to revolutionize agriculture through integration with nanoformulated fertilizers and pesticides [51]. Furthermore, the integration of Internet of Things (IoT) platforms enables real-time soil monitoring via cloud-connected sensor networks, allowing farmers to remotely optimize fertilization and irrigation strategies.

To better understand the diversity and functionality of these technologies, the following subsections outline key sensor types currently employed in soil nutrient monitoring:

2.5.1 Colorimetric and optical techniques

Colorimetric and optical sensor technologies have advanced rapidly, enabling non-destructive analysis of soil nutrients by measuring the interaction of light with soil surfaces. These systems commonly employ light-emitting diodes (LEDs) and photodiodes

to detect specific wavelengths absorbed by soil constituents, translating reflectance data into digital signals for classifying nutrient levels as high, medium, or low. For example, Masrie et al. [117] developed a sensor using three LEDs, each tuned to the absorption band of nitrogen (N), phosphorus (P), or potassium (K), and a photodiode to measure reflected light. Lower reflectance indicated higher nutrient absorption, allowing qualitative estimation of NPK concentrations. Compact colorimeters fabricated using MEMS technology have enabled accurate, low-cost field testing with minimal sample volumes. Yamin et al. [204] demonstrated a digital soil test kit for oil palm farms that achieved high accuracy 91.7% for nitrogen, 89.6% for phosphorus, and 93.8% for potassium closely matching laboratory results. Widely available colorimetric test kits offer affordable per-test costs and utilize established reagents such as diazotize dye for nitrate (NO_3^-), Nessler reagent for ammonium (NH_4^+), stannous chloride for phosphorus, tetraphenyl boron for potassium, and EDTA titration for calcium and magnesium [47]. Nutrient concentrations are typically determined by comparing the developed color to a reference strip, though this method is subject to human interpretation and limited resolution. To improve accuracy, recent approaches incorporate photometers and image processing. Nuntaporn et al. [137] introduced a portable colorimetric analyzer using a smartphone camera for phosphorus detection, enhancing field applicability. Despite their utility, colorimetry-based techniques often require multiple sample preparation steps, which can increase time and labor demands.

2.5.2 Spectroscopy-based sensors

Spectroscopy-based techniques offer rapid, precise soil nutrient analysis with minimal sample preparation compared to colorimetric methods. A range of spectroscopic modalities including visible (Vis), ultraviolet (UV), infrared (IR), X-ray fluorescence (XRF), inductively coupled plasma (ICP) spectroscopy, Laser-Induced Breakdown Spectroscopy (LIBS), and Near-Infrared Reflectance Spectroscopy (NIRS) have been widely adopted for laboratory-based soil testing [94]. Vis and UV spectrophotometers operate on the principle that electrons in atomic orbitals absorb photons of specific energies, producing characteristic absorption spectra. LIBS determines nutrient concentrations by analyzing the emission spectra from laser-induced plasma generated on the sample surface, while NIRS evaluates light absorbance and reflectance patterns based on soil composition. Both techniques are considered accurate, cost-effective, and operationally straightforward compared to conventional chemical assays. Monteiro et al. [124] demonstrated the use of modular UV–Vis sensors for analyzing hydroponic fertilizer, showcasing the adaptability of spectroscopic tools in nutrient monitoring. In IR spectroscopy, samples are exposed to infrared radiation, and molecular bonds absorb frequencies corresponding to their vibrational modes. The resulting spectrum, recorded by a spectrophotometer, reveals distinct absorbance bands associated with specific chemical bonds. Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) involves placing the test sample within an argon plasma core heated to approximately 8000 °C. This thermal excitation causes elements to emit light at characteristic wavelengths. A spectrometer scans the emitted light, and the amplitude at each wavelength is used to quantify elemental concentrations [75]. XRF spectroscopy operates on the principle that primary X-rays or electron bombardment excite atoms in the sample, causing them to emit secondary X-rays (fluorescence). Each element emits X-rays with a unique wavelength

and intensity, which are inversely proportional to the square of its atomic number and directly related to its concentration [135].

2.5.3 Ion-selective electrodes (ISEs) for N, P, K detection

Ion-selective electrodes (ISEs) and ion-selective field-effect transistors (ISFETs) are among the most widely used electrochemical sensing technologies for detecting inorganic ions in soil. Their selectivity is governed by ion-selective membranes (ISMs), which fall into three primary categories: (i) solid-state membranes composed of inorganic salt crystals; (ii) glass membranes, primarily used for pH measurement; and (iii) polymer membranes embedded with ionophores [42]. Polymer-based ISMs are typically formulated by combining a high molecular weight polymer matrix (for ionophore immobilization), a plasticizer (to enhance membrane flexibility and facilitate ionophore dissolution), and an ion exchanger (to improve selectivity for specific anions or cations). Altammar [12] detailed the synthesis of ISMs tailored for the detection of various ionic species, highlighting their adaptability for soil nutrient sensing. The detection mechanism of an ISE is based on potentiometric electrochemical principles. Unlike redox-based sensors, ISEs generate an electric potential when the target ion selectively binds to the ISM. The sensor comprises a complete galvanic cell with two electrodes: a working electrode (WE) and a reference electrode (RE). The WE is coated with an ISM specific to the target ion, and the potential difference between the WE and RE reflects the ion's activity or concentration in the test solution [74]. Table 2 describes comparative analysis of methodologies and performance for soil macronutrient and micronutrient detection. Figure 3a represents the schematic of a conventional ISE and Fig. 3b represents the schematic of a planar all solid-state ISE.

2.5.4 Conductivity measurement method for soil NPK analysis

EC measurement is a rapid and cost-effective technique for estimating soil nutrient levels, particularly nitrogen (N), phosphorus (P), and potassium (K). The method involves inserting two or three electrodes commonly made of steel, silver, or platinum into the soil sample and applying an alternating current (AC) voltage. Ion-driven current changes are detected using a multimeter, and the resulting EC values are correlated with nutrient concentrations. Darmawan et al. [44] explored the relationship between soil EC, capacitance, and NPK levels. Their findings indicated that while capacitance exhibited greater sensitivity in moist soil ($R^2 = 0.55$ vs. 0.34 for EC), conductivity showed a stronger linear correlation with NPK concentrations in dry soil ($R^2 = 0.81$). Given that soil moisture significantly influences EC readings, integrating additional electrical parameters may enhance the accuracy of nutrient estimation. Othaman et al. [141] reported a direct correlation between EC and nutrient/salt concentrations, alongside an inverse relationship with soil depth. EC and total dissolved solids (TDS) measurements support fertilizer requirement assessments, making this method suitable for large-scale soil testing due to its speed and affordability. However, its accuracy can be compromised by confounding factors such as pH and organic matter content, limiting its reliability for in-field diagnostics. Ultimately, selecting the appropriate nutrient assessment method depends on user expertise, soil and crop characteristics, available resources, and the required level of precision. EC-based techniques serve as valuable preliminary tools to identify zones requiring further analysis or targeted nutrient management.

Table 2 Comparative analysis of methodologies and performance for soil macronutrient and micronutrient detection

Technology used	Principle	Target nutrients	Advantages	Disadvantages	References
Ion-selective electrodes (ISEs)	Electrochemical sensors with ion-selective membranes	Nitrate (NO_3^-), Ammonium (NH_4^+), Potassium (K^+)	Real-time measurement, portable, more realistic pricing, high sensitivity	Interference from other ions; requires frequent calibration; short lifespan	[14, 93]
Optical/colorimetric	Reflection and absorption of light, Beer-Lambert's law	NO_3, NH_4, P, K, and micronutrients like Ca, Mg	Cost-effective, easy to use (available in in kit form); high volume in lab	Measurements are not direct, tedious sample preparation, not so accurate for field	[118]
Spectroscopy (NIR/MIR)	Analyzes absorption and emission of light at specific wavelengths	Total N, Organic Matter	Faster, Accurate, non-destructive; no chemical reagents needed; suitable for field use	Indirect measurement; requires extensive calibration models; high initial setup cost	[174, 175]
LIBS (laser-induced breakdown spectroscopy)	Analyzes atomic emission lines by creating a laser-induced plasma on the soil surface	P, K, Ca, Mg, and micronutrients	Rapid, multi-elemental analysis; minimized sample preparation; suited for field use	High setup cost; requires advanced data analysis; less effective for nitrogen	[87]
Nanomaterial-based biosensors	Use nanomaterials (e.g., graphene, nanoparticles) to enhance electrochemical or optical signals	Nitrate, Phosphate	Better sensitivity and selectivity; potential for miniaturization	Research and development are in progress; long-term stability has not yet been proven	[97]

2.5.5 MEMS based sensors

Micro-Electro-Mechanical Systems (MEMS) technology has revolutionized soil monitoring by enabling the development of compact, high-performance sensors for compositional analysis. MEMS sensors are characterized by their low mass and high mechanical resonance frequencies, which enhance sensitivity and responsiveness in field conditions. Their miniaturized form factor also provides a high surface-to-volume ratio, making them particularly effective for surface-based sensing mechanisms. A notable advancement in MEMS-based soil sensing is the work by Dudala et al. [51], who developed a low-cost, fully integrated microfluidic device for rapid nitrite detection. This point-of-source system utilizes Griess assays within a polydimethylsiloxane (PDMS) microfluidic platform composed of two distinct molds. Nitrite detection is performed spectrophotometrically, with the system incorporating a microcontroller for operational control, a syringe-pump-based flow injection system driven by a stepper motor, and a printed circuit board (PCB) to amplify photodiode output. Despite their promise, potentiometric ion-selective sensors face several challenges in soil applications. These sensors rely on solution fillers to detect specific ions, but they are prone to output potential drift over time, complicating continuous monitoring and field deployment [3]. Environmental factors such as temperature and pressure fluctuations can cause evaporation of the internal

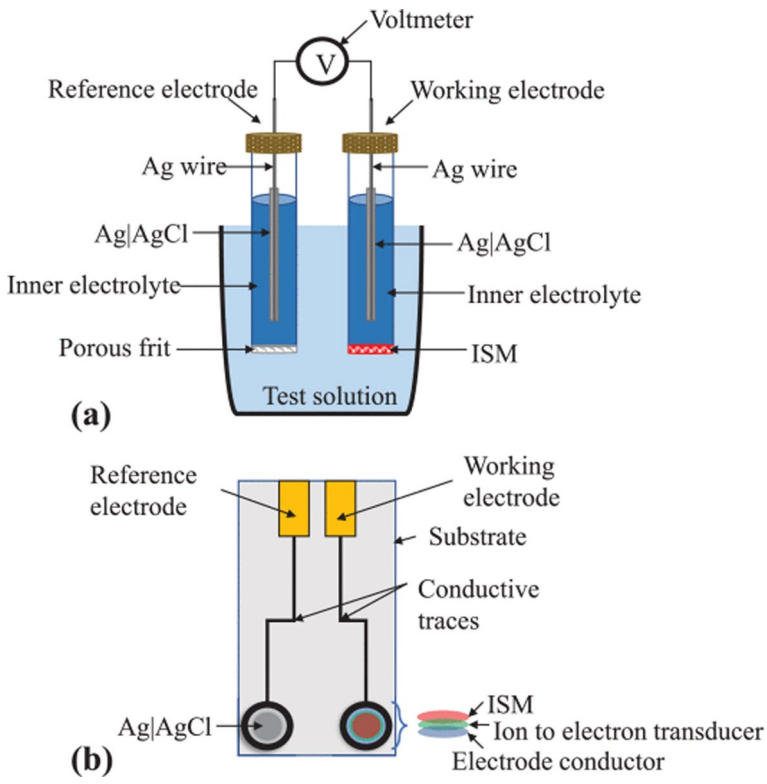

Fig. 3 **a** Schematic of a conventional ISE. **b** Schematic of a planar all solid-state ISE

filling solution, while particulate matter may obstruct microchannels or actuators, further hindering real-time performance.

2.5.6 Polymer-based sensors

Polymer-based sensors offer a cost-effective and versatile approach to soil nutrient monitoring, owing to the wide availability of functional polymers and simplified fabrication processes. The unique chemical and physical responses of different polymers to specific soil nutrients enable targeted detection of both macro- and micronutrients. Advanced polymers provide a flexible platform for sensor development. Mahmud et al. [143] reported a polymer-based sensor utilizing poly(3,4-ethylenedioxythiophene) (PEDOT) in conjunction with conductivity measurement to detect nitrate concentrations ranging from 1 to 1000 ppm within less than one minute. This rapid response highlights the potential of conductive polymers for real-time nutrient sensing. Al-Jeda et al. [11] developed a comprehensive polymer sensor system capable of monitoring environmental parameters such as temperature, humidity, pH, and soil moisture, alongside nutrient concentrations. Their work incorporated photocurable polyurethane-based polymers for potassium detection and iron oxide-polypyrrole nanocomposites for gaseous nitrogen sensing, demonstrating the adaptability of polymer matrices for multi-analyte detection.

2.5.7 Nanomaterial-based sensors

The integration of nanomaterials has markedly improved the sensitivity, selectivity, and responsiveness of soil nutrient sensors. Materials such as carbon nanotubes, graphene, and metal nanoparticles have been incorporated into sensor architectures to lower detection limits and accelerate response times [97]. Recent advancements have focused

on nitrate and ammonium sensors (Fig. 4), where nanomaterial-based platforms outperform conventional technologies in terms of precision and reliability. In agricultural applications, nanotechnology has facilitated the development of agrochemical-encapsulated nano-fertilizers, nano-carrier systems, nano-pesticides, nano-plant growth promoters, and nano-herbicides, offering controlled release and enhanced bioavailability [81]. Nanobiosensors, in particular, have demonstrated superior sensitivity and dependability compared to traditional biosensors, enabling detection of analytes at significantly lower concentrations [73]. A notable example is the nanobiosensor developed by Khaledian et al. [99] for field detection of *Ralstonia solanacearum* DNA at the unamplified stage. This sensor utilized single-stranded DNA probes conjugated to gold nanoparticles. Upon acid treatment, successful hybridization with target DNA maintained the solution's red colour, indicating nanoparticle stability. In contrast, failed hybridization led to nanoparticle aggregation and a visible colour shift to purple. This method provided a rapid, simple, and visual approach for pathogen identification. The enhanced performance of nanomaterial-based sensors is attributed to their atomic-scale operation, high surface-to-volume ratio, elevated sensitivity, and minimal error rates. However, nanobiosensors remain in the early stages of development and require further refinement for widespread field deployment. Their potential to transform agricultural diagnostics and nutrient management is substantial, with ongoing research expected to expand their applicability [161].

2.5.8 Multi-parameter NPK sensors

Recent commercial advancements have led to the development of integrated sensor systems capable of simultaneously measuring N, P, and K concentrations in soil. These multi-parameter sensors typically utilize electrochemical detection mechanisms, incorporating multiple working electrodes each selectively optimized for a specific nutrient ion. To enhance accuracy and adaptability across diverse soil types, machine learning algorithms have been integrated into these systems. These algorithms facilitate dynamic calibration and compensate for soil matrix effects, improving the reliability of nutrient

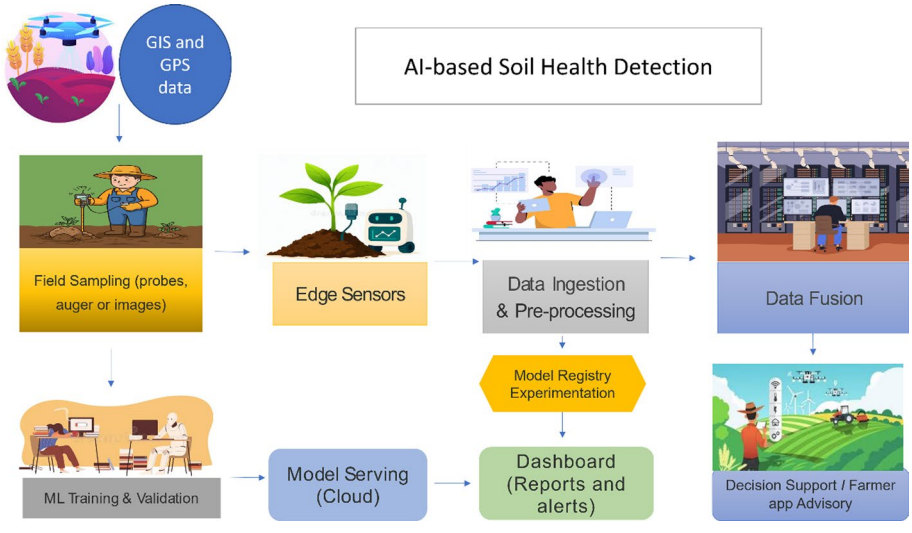

Fig. 4 Flow diagram illustrating an AI-based soil health detection system

quantification under variable field conditions [101]. Such innovations mark a significant step toward real-time, data-driven nutrient management in precision agriculture.

2.6 Temperature sensors

Temperature plays a pivotal role in agricultural processes beyond photosynthesis, influencing seed germination, flowering, grain development, and overall plant growth. Most physiological activities in plants are optimized within a temperature range of –10 to 50 °C. Additionally, temperature affects the physico-chemical and microbiological dynamics of soil, which in turn modulate nutrient availability and plant health [34]. Historically, soil temperature was measured using mercury-in-glass thermometers inserted to specific depths. However, modern advancements have introduced a range of precise and efficient technologies, including thermocouple-based sensors, thermistor-based sensors [105], semiconductor-based devices, resistance temperature detectors (RTDs) [45, 164], and infrared temperature sensors. These innovations have significantly improved the accuracy, reliability, and scalability of soil temperature monitoring. Fundamentally, soil temperature sensors operate on the principle of the thermoelectric effect. They comprise a temperature-sensing element (e.g., thermocouples or probes) that converts thermal variations into electrical signals [158]. These signals are processed through a data acquisition circuit, which digitizes the input, and a data processing unit that enables interpretation and visualization for end users [207]. To accommodate the diverse requirements of agricultural environments, several types of temperature sensors have been developed, each offering distinct advantages in terms of sensitivity, durability, and integration potential.

2.6.1 Thermocouple-based

Thermocouples operate on the principle of the Seebeck effect, wherein a junction of two dissimilar metals generates a small voltage in response to a temperature gradient. When the junction is heated or cooled, the resulting thermoelectric circuit produces a differential voltage, which correlates directly with the temperature variation. This voltage can be measured and interpreted to determine soil temperature with high accuracy. Thermocouple-based sensors are widely recognized for their rapid response time, cost-effectiveness, and reliability in field conditions. Furthermore, the integration of wireless thermocouples facilitates remote temperature monitoring, eliminating the need for physical cabling and enhancing scalability in precision agriculture applications [90].

2.6.2 Thermistor-based

A thermistor is a type of temperature sensor composed of a semiconductor material whose electrical resistance varies predictably with temperature. As the ambient soil temperature changes, the thermistor's resistance shifts accordingly, leading to a corresponding change in the electrical current when a constant voltage is applied. This variation in current is then interpreted by a signal conditioning circuit or digital meter, which converts it into a temperature reading. Thermistors are particularly valued for their high sensitivity, compact size, and cost-effectiveness, making them suitable for localized soil temperature monitoring in agricultural applications.

2.6.3 Resistance temperature detectors (RTDs)

Resistance Temperature Detectors (RTDs) measure soil temperature by monitoring the voltage drop across a metal resistor while a constant current is applied [199]. The sensing element typically consists of a fine metal wire, such as platinum, coiled around a ceramic or glass core, ensuring mechanical stability and thermal responsiveness. RTDs exhibit a linear and repeatable change in resistance with temperature, allowing for precise and stable measurements. Compared to thermocouples, RTDs offer higher accuracy, better long-term stability, and lower signal drift, although they may be more sensitive to mechanical strain and require careful calibration [125]. These sensors are particularly suited for surface temperature monitoring.

2.6.4 Integration with moisture sensors

Modern soil monitoring systems increasingly integrate temperature and moisture sensors to support advanced modeling of evapotranspiration, root zone hydrodynamics, and soil water balance [212]. By capturing simultaneous data streams, these systems enable a more comprehensive understanding of soil thermal regimes, heat flux, and hydrothermal interactions, which are critical for optimizing irrigation strategies and predicting crop responses under variable climatic conditions. Such integrated platforms also facilitate real-time decision-making in precision agriculture by enhancing the spatial and temporal resolution of soil property assessments [132, 195].

2.6.5 Multi-depth temperature profiling

Contemporary soil temperature sensing systems increasingly employ multi-depth sensor arrays to capture temperature variations across vertical soil profiles [83, 84]. By embedding sensors at strategic depths, these systems generate high-resolution thermal data that reveal soil heat flux, thermal conductivity, and diurnal temperature gradients. Such profiling is essential for modeling soil thermal dynamics, which directly influence seed germination timing, root development, and microbial activity [206]. This approach enhances agronomic decision-making by aligning planting schedules with optimal thermal conditions and improving predictions of crop emergence under fluctuating climate scenarios.

2.6.6 Optical fibre-based temperature sensors

Conventional fibre-optic temperature sensors typically utilize silica-based fibres, which exhibit limited temperature sensitivity often less than 0.3 nm/°C due to the uniform thermal response of both the core and cladding materials [68]. To address this constraint, Fakhruldeen et al. [59] proposed the use of Photonic Crystal Fibres (PCFs), a subclass of Microstructured Optical Fibres (MOFs), selectively filled with thermosensitive liquids such as alcohol. The temperature-dependent refractive index of the liquid significantly alters the fibre's light-guiding characteristics, enhancing sensitivity. In their design, a highly birefringent alcohol-filled PCF was embedded within a temperature-controlled fibre loop mirror, achieving a remarkable sensitivity of 6.6 nm/°C, far exceeding that of traditional fibre sensors [107]. This advancement opens new possibilities for high-resolution, passive temperature monitoring in soil and environmental applications, particularly where electromagnetic interference or harsh conditions limit the use of electronic sensors.

2.7 Biological sensors (biosensors)

A biosensor is an analytical device that combines a biological recognition element such as an enzyme, antibody, or nucleic acid with a physicochemical transducer to produce a measurable signal in response to the presence of a specific analyte [79]. These sensors are capable of detecting and quantifying minute concentrations of biological markers, pathogens, toxic chemicals, and environmental parameters such as pH. The electronic component of the biosensor processes the signal generated by the bioreceptor, enabling real-time monitoring, data transmission, and decision support in field applications. Biosensors are available in a variety of sizes, shapes, and configurations, tailored for specific detection environments. A conventional biosensor typically comprises the following components: (i) Analyte—the target substance to be detected, (ii) Bioreceptor—the biological element that interacts with the analyte, (iii) Transducer—converts the biorecognition event into a measurable signal; electronics—amplifies and processes the signal and display unit—presents the output to the end-user. These devices are increasingly used in agricultural diagnostics, soil health monitoring, and environmental surveillance, offering high sensitivity, specificity, and portability.

2.7.1 Microbial activity detection

A multi-biosensor chip capable of detecting harmful algal species in field conditions was developed by Diercks et al. [46]. This innovative platform utilized an array of 16 gold electrodes, enabling the simultaneous identification of up to 14 target species, thereby facilitating high-throughput environmental diagnostics. Building on this multiplexing approach, McCoy et al. [119] introduced a surface plasmon resonance biosensor designed to monitor blooms of Alexandrium minutum along coastal zones. The biosensor's performance showed strong correlation with traditional light microscopy, validating its accuracy and field applicability. Further advancements in microbial detection include the electrochemical DNA biosensor proposed by Toldra et al. [184] for identifying *Ostreopsis cf. ovata* in both planktonic and benthic samples. This biosensor demonstrated detection results comparable to qPCR and colorimetric assays, highlighting its potential for rapid, cost-effective monitoring of microbial contaminants. In terrestrial ecosystems, modern biosensors are increasingly employed to assess microbial activity by detecting specific enzymes and metabolic by-products [83, 84]. Techniques such as fluorescent labeling and electrochemical transduction are commonly used to quantify key microbial processes, including respiration, nitrogen cycling, and organic matter decomposition [195]. These capabilities are critical for evaluating soil health, nutrient turnover, and carbon dynamics, thereby supporting sustainable agricultural management and ecological modeling.

2.7.2 Aptamers- DNA/RNA-based sensors

Aptamers are synthetic, single-stranded DNA or RNA sequences that fold into specific 2D or 3D conformations, enabling high-affinity and selective binding to target molecules. Their high surface density and reduced spatial hindrance enhance binding efficiency, making them ideal for biosensing applications. Aptamers exhibit remarkable thermal refolding capabilities and remain stable across a wide range of temperatures, pH levels, and storage conditions [49]. Unlike antibodies, aptamers are chemically synthesized, cost-effective, and easily tailored to meet specific detection requirements. The

selection of aptamers is typically performed using Systematic Evolution of Ligands by Exponential Enrichment (SELEX), along with its advanced variants such as cell-SELEX, capillary electrophoresis-SELEX, and others. Aptamer-based biosensors, commonly referred to as aptasensors, employ diverse transduction mechanisms including optical, electrochemical, and piezoelectric approaches. These can be further classified into following sections: (i) Labeled formats utilizing fluorescent dyes (e.g., fluorescein) for signal generation, and (ii) Label-free formats employing techniques such as surface plasmon resonance (SPR) for direct detection. Furthermore, recent innovations have expanded the field applicability of aptasensors through the integration of portable PCR systems and loop-mediated isothermal amplification (LAMP) technologies, enabling rapid, on-site molecular diagnostics [27].

2.7.3 Enzyme activity sensors

Enzymes are highly efficient biocatalysts that accelerate biochemical reactions through specific catalytic binding, making them ideal components in biosensors for detecting target analytes. Enzyme-based biosensors operate through three primary mechanisms of analyte recognition: (i) Enzymatic metabolism of the analyte, where the sensor measures the catalytic transformation of the substrate. (ii) Inhibition or activation of enzymatic activity, which correlates with changes in product synthesis. (iii) Alterations in enzyme properties, such as conformational shifts or electrochemical behavior, in response to analyte interaction. Due to their high specificity, enzyme-based biosensors are widely used in detecting compounds such as glucose, urea, and lactate. However, their structural fragility poses challenges in enhancing sensitivity, stability, and environmental adaptability, particularly under variable field conditions. These biosensors commonly employ electrochemical transducers, which convert biochemical reactions into measurable electrical signals. Notable examples include a potentiometric urea biosensor with a 30-s response time and a detection limit of 0.77 μM [133], and W-Au-based amperometric glucose biosensors designed for real-time brain glucose monitoring [41]. Recent advancements involve the integration of enzymes with nanomaterials, which enhances signal transduction, enzyme stability, and sensor miniaturization, thereby expanding their applicability in both clinical and environmental diagnostics.

2.7.4 Nanoparticle-based biosensors

Nanoparticle-based biosensors represent a cutting-edge advancement in biosensor design. Nanomaterials (NMs) have recently been recognized as both bioreceptors and transducers due to their unique physicochemical properties. Cerium oxide-based NMs, for example, exhibit biomimetic catalytic activity that enhances analyte recognition [170]. Inorganic nanomaterials such as noble metal nanoparticles, quantum dots (QDs), graphene, and carbon nanotubes have demonstrated exceptional transduction capabilities, offering high conductivity, surface area, and signal amplification. These properties have enabled the development of highly sensitive, miniaturized biosensors suitable for multiplexed and real-time environmental monitoring [67].

The evidence presented throughout this section collectively demonstrates a resounding affirmative answer to RQ1. Advancements in material science and transducer design have fundamentally enhanced soil sensor performance. We have documented the evolution from mechanical tensiometers and fragile glass electrodes to solid-state

ISFETs, dielectric probes with sophisticated electronics, and optical sensors employing novel dyes and nanomaterials. These innovations have directly resulted in marked improvements in accuracy through reduced signal drift and better temperature compensation, increased specificity via ion-selective membranes and spectral fingerprinting, and greater field-deployability afforded by miniaturization, wireless connectivity, and reduced power demands. The integration of MEMS technology, conductive polymers, and nanobiosensors represents a paradigm shift from mere measurement devices towards intelligent, multi-analyte sensing platforms. Therefore, we conclude that interdisciplinary progress at the intersection of materials engineering, electrochemistry, and photonics has been the principal catalyst in transitioning soil health monitoring from a periodic, lab-based exercise to a dynamic, in-situ, and data-rich process essential for modern sustainable agriculture.

2.7.5 Soil gas sensors

Soil gas sensors can detect and monitor gases such as oxygen, carbon dioxide, methane, and nitrous oxide in the soil environment. These sensors play a crucial role in studying soil respiration, root activity, microbial processes, and other vital soil ecological processes. It generally works on the principle of electrochemical or optical methods, where the gas is allowed to react with a sensitive element to produce a signal that is measured by these sensors. A sensor chamber is placed at the soil site to capture gases, that is detected by the sensor element. The data logger or microcontroller records this data, sends it to the communication modules for automated sampling. Based on working principle, soil gas sensors can be classified as electrochemical, optical, infrared (NDIR), catalytic and metal oxide (semiconductor). They can also be categorized in to different types, based on the gases they can detect, such as CO_2, O_2, NH_3, H_2S, CO, etc.

Advancements in soil gas sensors will offer a real-time, in-situ, and cost-effective measure for quantifying soil gas movements. Reference [60] forming an integral part of precision agriculture, carbon accounting, and environmental protection initiatives. By use of chemical or physical interactions between the gas molecules and the sensor surface, soil gas sensors can help identify variations in the concentration of the gas. Redox interactions between gas molecules and the electrode surface produce detectable currents that are proportionate to the concentration of the gas [120].

Out of the varying gas sensors, NDIR-based CO_2 sensors (e.g., Vaisala CARBOCAP) are widely used due to their stability and selectivity. Miniaturized IR sensors have been developed for field use [203]. Cascade laser (QCL) sensors also offer accurate and selective detection [43] but their use is limited to higher cost. Integrated systems using wireless sensor networks (WSN) or Internet of Things (IoT) platforms enable continuous multi-gas monitoring. These systems combine microcontrollers, data loggers, and remote telemetry for spatial mapping of soil gas fluxes [131].

Roscioli et al. [157] described a method to measure soil gas using an approach that uses specially designed PTFE (a durable, non-reactive plastic) probes to sample gases directly from the soil with minimal disturbance. These probes collect trace gases produced by soil microbes and send them to sensitive analyzers above ground, where concentrations and isotopic signatures can be measured in real time. The entire system enables researchers to observe how microbial activity in the soil responds to environmental changes (like fertilizer addition or shifts between aerobic and anaerobic conditions) with

high spatial and temporal resolution, providing a much deeper understanding of soil biochemical cycles and "hot" zones of activity that might otherwise be missed.

A portable MOS (metal oxide semiconductor) gas sensor-type electronic-nose device (the PEN3 e-nose) was evaluated to differentiate between healthy and inoculated root fragments [19]. In urban environments, fungal sapwood decays are very common and also the most damaging diseases of trees, resulting in serious damage and losses. The metal oxide semiconductor present in this device can easily detect the volatile compounds in the air, and it memorises each aroma or chemical, creating a 'fingerprint' for each compound identified. Electronic nose has also been used for detecting different types of rots, such as in potato brown rot and ring rot [29].

Romanak et al. [156] proposed another process-based method for detecting CO_2 leakage in vadose zones overlying deep geological storage sites. Conventional methods often require extensive background monitoring and is generally complicated due to natural soil CO_2 variation. On other hand, this approach analyzes chemical relationships among multiple soil gases such as N_2, O_2, CO_2, and CH_4 for immediate distinction of leakage signatures from signals caused by natural microbial or geochemical processes. By examining patterns such as O_2 versus CO_2 and CO_2 versus N_2/O_2, this method identifies anomalous gas fingerprints indicative of deep-sourced CO_2 or methane intrusion. The combination of nanomaterials, artificial intelligence (AI), and Internet of Things (IoT) technologies is vital to soil gas monitoring. In the field, machine learning methods can enhance signal calibration and interpretation [140]. High-resolution soil gas mapping will be made possible by combining sensor networks with remote sensing and modeling, assisting with climate mitigation and sustainable land management techniques. Because they provide real-time insights into soil processes and gas fluxes, soil gas sensors represent a paradigm shift in environmental monitoring. To get above current constraints, more research is needed in the areas of data integration, sensor shrinking, and material innovation. Soil gas sensors will be essential to the shift to carbon–neutral farming and sustainable ecosystems as climate regulations and agriculture demand more precise GHG monitoring.

3 Emerging technologies in soil sensing

To address our second research question (RQ2), this section critically examines the paradigm shift from discrete sensor units to interconnected sensing systems. We will evaluate the transformative potential of Wireless Sensor Networks (WSNs), the Internet of Things (IoT), and advanced spectroscopy as synergistic platforms that collectively transcend the constraints of traditional methods. Our analysis will focus on their capacity to provide high-resolution spatial coverage across entire fields and enable continuous, real-time temporal monitoring, thereby moving beyond sparse, periodic manual measurements. We will dissect how these platforms facilitate data fusion, enhance spatial interpolation, and enable predictive analytics, ultimately creating a dynamic, multi-scale understanding of soil heterogeneity. This assessment will also rigorously consider the persistent challenges such as connectivity, energy autonomy, data processing burdens, and calibration complexities that currently define the practical limits of their extensibility and adoption in diverse agricultural landscapes. A comprehensive evaluation of integrated sensing platforms presented in Table 3.

Table 3 Capabilities and limitations of integrated sensing platforms for overcoming spatial and temporal constraints in soil monitoring

Integrated platform	Core function and scale	Key technological enablers	Advantages over point sensing (addressing RQ2)	Persistent challenges and limitations	References
Wireless sensor networks (WSNs)	Field scale: dense, in-situ network for continuous multi-parameter data logging	Low-power sensors (e.g., capacitive FDR), Zigbee/LoRa protocols, mesh networking, solar harvesting	Temporal: real-time, continuous data streams. Spatial: denser grid of measurement points enables accurate spatial interpolation and variability mapping	Limited node battery life; signal attenuation in dense crops; high initial deployment cost for full-field coverage; requires robust gateway infrastructure	[139, 154, 213, 214]
Internet of things (IoT) systems	Farm-to-cloud scale: automated data acquisition, cloud storage, and predictive analytics	LPWAN (e.g., LoRaWAN, NB-IoT), cloud computing, ML algorithms, mobile app interfaces	Temporal: enables closed-loop automation (e.g., irrigation). Spatial: integrates data from multiple fields for unified management. Facilitates predictive spatial modeling of trends (e.g., salinity)	Cellular/Wi-Fi dependency in remote areas; data security and ownership concerns; requires high level of technical expertise for setup and maintenance	[17, 18, 56, 95, 96]
Portable spectrometers (NIR/MIR)	Point-to-plot scale: rapid, on-site proximal sensing of multiple soil properties	MEMS-based optics, LED light sources, on-board processing, Bluetooth connectivity	Temporal: provides immediate results, eliminating lab turnaround time. Spatial: allows for high-density, guided grid sampling across a field to build accurate spatial maps	Requires extensive and soil-specific calibration; performance sensitive to soil moisture and surface conditions; measures only surface properties	[31, 32, 129, 130, 174, 175]
UAV-based multispectral/Hyperspectral imaging	Field scale: high-resolution (cm-dm) spatial mapping of surface soil and crop indicators	Lightweight sensors, GPS-guided autonomous flight, photogrammetry, machine learning for image analysis	Spatial: extremely high spatial resolution captures fine-grained variability within fields. Temporal: on-demand deployment allows for monitoring at critical phenological stages	Limited to surface observations; data volume requires significant processing; cost of UAV and sensor payload; flight time and regulatory limitations	[8, 28, 159]
Satellite-based remote sensing	Regional scale: synoptic, repeated coverage for large-area monitoring and trend analysis	Multi-spectral sensors (e.g., Sentinel-2, Landsat), synthetic aperture radar (SAR, e.g., Sentinel-1), public data access	Spatial: unmatched spatial extent for regional studies. Temporal: long-term archives (decades) enable historical analysis and time-series trend monitoring	Coarse spatial resolution (10 m-1 km) obscures field-level variability; infrequent revisit time (days-weeks); signals attenuated by atmosphere and vegetation canopy	[126, 195]
AI/ML data fusion platforms	Multi-scale: synthesizes data from all above platforms into predictive models	Cloud computing, deep learning (CNNs, RNNs), geostatistical algorithms, digital twin frameworks	Spatial/temporal: translates sparse point data into continuous forecasted maps (e.g., soil moisture prediction); fuses high-res UAV with frequent satellite data for optimized spatiotemporal resolution	"Black box" models can lack transparency; requires large, high-quality datasets for training; computational intensity and expertise required	[108, 148, 192, 194]

3.1 Wireless sensor networks (WSN)

Wireless Sensor Networks (WSNs) consist of spatially distributed sensor nodes that wirelessly collect and transmit data on key soil parameters such as temperature, moisture content, pH, and nutrient levels [186]. These networks enable real-time remote monitoring across large agricultural landscapes, significantly reducing the need for manual intervention and enhancing data-driven decision-making [103]. WSNs are particularly advantageous for large-scale field data acquisition, as they require minimal physical infrastructure while offering extensive spatial coverage [139]. For instance, their deployment in berry orchards has led to improved irrigation efficiency and microclimate monitoring through continuous soil moisture tracking [165]. When integrated with Geographic Information Systems (GIS), WSNs facilitate the mapping of soil variability, enabling precision resource allocation such as targeted fertilization and irrigation [126].

The incorporation of cloud computing further enhances WSN functionality by providing scalable data storage, real-time access via web or mobile platforms, and advanced analytics for soil health assessment [197]. This integration supports predictive modeling for irrigation scheduling and nutrient management, thereby boosting agricultural productivity [95, 96]. A notable application includes the use of WSNs in vineyards, where cloud-based analytics have enabled accurate tracking of soil moisture trends for optimized water management [213, 214]. Despite their benefits, WSNs face several operational challenges. These include limited battery life, signal interference in densely vegetated areas, and the need for robust network protocols to ensure reliable data transmission [154]. Ongoing research focuses on energy-efficient designs, such as solar-powered sensor nodes, and the development of advanced routing algorithms to improve network resilience and performance in agricultural environments [139].

3.2 Internet of Things (IoT) integration

The Internet of Things (IoT) has become a cornerstone of precision agriculture by enabling the automated collection, transmission, and analysis of soil data through interconnected sensor networks [17, 18]. IoT-based sensor nodes transmit real-time information on soil parameters such as moisture, temperature, pH, and nutrient levels to cloud platforms using communication protocols like Wi-Fi or LoRaWAN. The latter is particularly suited for remote agricultural regions due to its long-range, low-power characteristics [35, 134]. In arid environments, IoT-enabled soil moisture sensors have supported automated irrigation systems, optimizing water use and reducing resource waste [165]. These systems are further enhanced by machine learning algorithms, which process large volumes of sensor data to model soil health trends, including moisture depletion, nutrient deficiencies, and salinity fluctuations [108]. Integration with multispectral drone imagery and AI-driven analytics enables the detection of water stress, canopy dynamics, and other physiological indicators, facilitating precision interventions such as targeted irrigation and nutrient delivery particularly in high-value crops like vineyards [123]. Predictive models derived from IoT data have proven effective in forecasting soil degradation risks and irrigation requirements, thereby improving agronomic decision-making [95, 96]. For example, machine learning enhanced irrigation scheduling in maize fields has led to measurable gains in yield efficiency [76]. The proliferation of user-friendly interfaces, such as mobile applications, has democratized access to IoT data, allowing farmers to monitor soil conditions in real time and respond proactively

[15]. Despite its transformative potential, IoT adoption in agriculture faces several challenges. These include high initial costs for sensor deployment, limited internet connectivity in rural areas, and the need for robust data security and device interoperability [56, 180]. To address these constraints, researchers are exploring cost-effective IoT architectures and hybrid connectivity models, such as combining satellite links with LoRaWAN, to ensure scalable and resilient deployment [35].

Imagery is processed with machine learning regressors to estimate soil nutrients and organic carbon. Liu et al. [110] used UAV hyperspectral imaging to create detailed nutrient maps across a field. Kumar et al. [104] combined topographic variables from digital elevation and vegetation/soil indices from Landsat to predict topsoil organic carbon (SOC) at a 30 m resolution using ensemble machine learning techniques, including Random Forest, XGBoost, and SVM. The best model, Random Forest, achieved a high calibration R2 of 0.966 and produced a 30 m SOC map for Northeast India. Other systems use ground-based sensor networks or geophysical measurements as predictors. Peng et al. [147] developed mlhrsm, an R package that combines in-situ moisture sensors, satellite-derived variables like vegetation and terrain, and existing soil moisture models. They used a quantile random forest to create daily soil volumetric water content maps at a 30 m resolution over a 70 ha cropland. Adhikari et al. [6] measured apparent electrical conductivity (a geophysical proxy) and topographic attributes on a 218-point grid in a Texas field. They then applied random forest to predict key soil properties such as CO_2 flux, organic carbon, organic nitrogen, and a composite Soil Health Index at a 5 m resolution. Key predictors included electrical conductivity, wetness index, valley flatness, and terrain position.

Smartphones and simple sensors enable AI-driven field tests. Ferreira da Silva et al. [63] set up a colorimetric paper test for soil pH. Farmers dip a reagent strip into soil extract and then photograph it with a smartphone app. A trained machine learning classifier uses the colorimetric pattern to categorize the soil pH as low, medium, or high. This mobile system operated under tropical field conditions and was tested against lab analysis. Folorunso et al. [64] created GeaGrow, a mobile app that employs artificial neural networks. They collected soil samples across southwest Nigeria and measured properties such as nitrogen, phosphorus, potassium, organic carbon, and pH. They trained the ANN to predict these properties based on input features like soil texture and water retention. The app offers farmers location-based recommendations for NPK and fertilizer.

Each approach requires data preprocessing and model training. For example, sensor and image data must be cleaned and normalized, and machine learning hyperparameters tuned. It is common to use ensemble methods or deep learning techniques such as Random Forest, XGBoost, SVM, and ANN. These models capture complex, nonlinear relationships between environmental variables and soil parameters. In summary, AI-based soil health detection includes the following steps: data acquisition (either remote or close), feature engineering (incorporating spectral indices, topography, etc.), model training and validation on soil lab samples, and spatial prediction or interpolation (Table 4).

Table 4 Comparative performance of AI/ML models for different parameters

Technique	Key strengths	Limitations	Soil parameters predicted	Accuracy/performance	Data Sources
Linear Regression (MLR)	Simple, interpretable; few data required and Low computational cost	Only linear relations (often underfits soil data) Sensitive to collinearity and outliers	pH, EC, basic nutrients (used as a baseline)	Typically, low–moderate (e.g. $R^2 \approx 0.5–0.7$)	[102]
Support Vector Machine (SVR)	Captures nonlinear patterns via kernels	• Scales poorly to very large datasets • Requires careful kernel selection	pH, EC, SOC, soil texture classes, nutrient levels	Often high ($R^2 \approx 0.7–0.9$ reported)	[102]
Random Forest (RF)	Ensemble of trees; handles noise and complex relationships	Less interpretable; can be computationally intensive	pH, EC, organic C, N, P, K, micronutrients (Zn, Fe,), texture, salinity	High (often $R^2 \approx 0.8–0.9$)	[102]
Gradient Boosting (XGBoost, etc.)	Very high accuracy via boosted trees Handles missing data, feature interactions	• Sensitive to hyperparameters; risk of overfitting • Training slower on big data	Similar targets as RF (pH, SOC, N, P, K, etc.)	Very high (e.g. XGBoost achieved $R^2 \approx 0.86$ for SOC	[205]
ANN (MLP)	• Flexible nonlinear modeling • Integrates diverse inputs (spectra, climate)	• Data-intensive; overfits without regularization • Requires tuning of architecture	pH, EC, OC, N, P, K, micronutrients, moisture	High when data are sufficient (often $R^2 > 0.8$)	[16]
Deep Neural Network (DNN)	• Multiple layers capture hierarchical features (state-of-art)	• Very data-hungry and computationally expensive	All above parameters at regional/global scales	Often outperforms shallow NN (e.g. 4–7% better SOC mapping vs. RF)	[65]
Recurrent NN (RNN/LSTM)	Captures temporal dynamics in sequential data	Complex; needs time-series data (rare for static soil)	Time-series moisture or nutrients in dynamic studies	High when temporal data available; used in hybrid CNN–RNN	[86]

3.3 Spectroscopy-based sensors

Spectroscopy-based sensors offer rapid and precise analysis of key soil parameters, including organic matter, texture, moisture, and nutrient content. These sensors primarily utilize Near-Infrared (NIR) and Mid-Infrared (MIR) wavelengths to characterize soil composition. While MIR spectroscopy excels in detecting mineral content and clay concentration, NIR spectroscopy is particularly effective for identifying organic molecules [145, 174, 175]. Recent advancements have significantly reduced reliance on laboratory-based soil analysis by enhancing the field-deployability of spectroscopic sensors [129, 130]. The development of portable spectrometers has made spectroscopy-based sensing more practical and affordable for on-site applications [69]. For example, handheld NIR sensors now enable farmers to conduct real-time assessments of soil organic carbon, facilitating informed decisions without the need for expensive lab testing [31, 32]. These tools have been successfully used to guide targeted fertilization based on localized soil conditions [191].

With some models priced below $1,000, these compact devices are increasingly accessible and user-friendly [129, 130]. However, due to the spatial variability of soil characteristics, accurate calibration for specific soil types remains essential [174, 175]. Current research focuses on developing universal calibration models and improving sensor resilience to environmental fluctuations, such as changes in moisture and temperature [136]. Integration with IoT platforms further enhances real-time analysis and decision-making capabilities [69]. Complementary technologies such as Microelectromechanical Systems (MEMS) have enabled the miniaturization of pH and electrical conductivity (EC) sensors, facilitating their inclusion in multi-parameter sensor platforms. These solid-state devices offer high precision, reduced production costs, and improved field applicability [97, 206]. In parallel, fluorescent nanoparticles notably quantum dots (QDs) have enabled real-time biological tracking. Aptamer-QD conjugates can detect specific targets such as proteins and cancer cells with high sensitivity. For instance, NIR PbS QDs have been used to detect thrombin protein within one minute, achieving a detection limit of approximately 1 nM [133]. Similarly, electrochemical impedance spectroscopy (EIS)-based biosensors employing gold nanoparticles (GNPs) have demonstrated biocompatibility and ultra-low detection limits for L-arginine, with a linear range of 0.1 pM to 0.1 mM and a detection threshold of 0.01 pM [62].

Despite their growing utility, spectroscopy-based sensors face several challenges. Calibration complexity remains a major hurdle, as soil heterogeneity particularly in texture, mineralogy, and moisture can significantly affect spectral readings [145, 174, 175]. This necessitates site-specific calibration models, which may limit scalability across diverse agroecological zones [129, 130]. Additionally, spectral interference from moisture, organic residues, or surface roughness can reduce measurement accuracy [31, 32]. While portable spectrometers have become more affordable, initial investment costs and technical training requirements may still pose barriers for smallholder adoption [69]. Environmental factors such as temperature fluctuations, dust accumulation, and sensor drift can further compromise long-term reliability [136]. Addressing these limitations requires continued innovation in sensor robustness, data preprocessing algorithms, and standardized calibration protocols to ensure consistent performance under field conditions [191].

3.4 Hyperspectral and multispectral imaging

Hyperspectral and multispectral imaging technologies, when deployed via drones or satellites, offer powerful tools for mapping soil variability and identifying areas of degradation [28]. Hyperspectral imaging captures data across hundreds of narrow spectral bands, enabling detailed characterization of soil properties such as texture, salinity, moisture, and organic matter [159]. In contrast, multispectral imaging, which utilizes fewer but broader bands, is more cost-effective and widely adopted for large-scale monitoring [8].

Drone-based hyperspectral photography facilitates high-resolution mapping of field-scale soil heterogeneity, supporting precision fertilization and irrigation control. Meanwhile, satellite-based multispectral imagery provides extensive spatial coverage, making it suitable for regional assessments such as desertification monitoring and soil health evaluation [8]. Recent innovations in lightweight sensors and extended drone battery life have improved accessibility for farmers and researchers alike. The vast volume of data

generated by hyperspectral systems necessitates sophisticated processing techniques, often involving machine learning algorithms to extract meaningful patterns from complex spectral signatures [28]. For instance, hyperspectral data combined with machine learning has significantly enhanced the accuracy of soil organic matter mapping [159]. These approaches enable predictive modeling and site-specific management strategies in precision agriculture.

Despite their potential, hyperspectral and multispectral imaging systems face several challenges. Soil-specific calibration requirements can limit generalizability across diverse landscapes, while atmospheric interference such as cloud cover and light scattering can distort satellite-based measurements [8]. Additionally, the high cost of hyperspectral sensors, complex data processing, and the need for specialized expertise may hinder widespread adoption. Ongoing research is focused on reducing hardware costs, streamlining data workflows, and developing standardized calibration models to support broader implementation in agricultural contexts (Fig. 5).

The evidence presented in this section demonstrates that integrated platforms can substantially, though not yet completely, overcome the historical spatial and temporal limitations of conventional soil sensing. WSNs and IoT systems have proven highly effective in establishing dense, wireless mesh networks that provide unprecedented temporal resolution and extensive spatial coverage, enabling real-time decision-making at the field scale. Advanced spectroscopic and imaging techniques, particularly when deployed on UAV and satellite platforms, have dramatically expanded our capacity for rapid, non-destructive spatial mapping of soil properties, effectively bridging the gap between point measurements and landscape-scale variability. However, our analysis also confirms that the full potential of these platforms is contingent upon overcoming significant hurdles related to energy sustainability, data management, and economic accessibility. Therefore, these integrated systems represent a monumental leap forward, their efficacy is ultimately a function of the supporting infrastructure and analytical frameworks, underscoring the need for continued innovation in edge computing, robust communication protocols, and scalable data fusion algorithms to achieve truly seamless and ubiquitous soil monitoring.

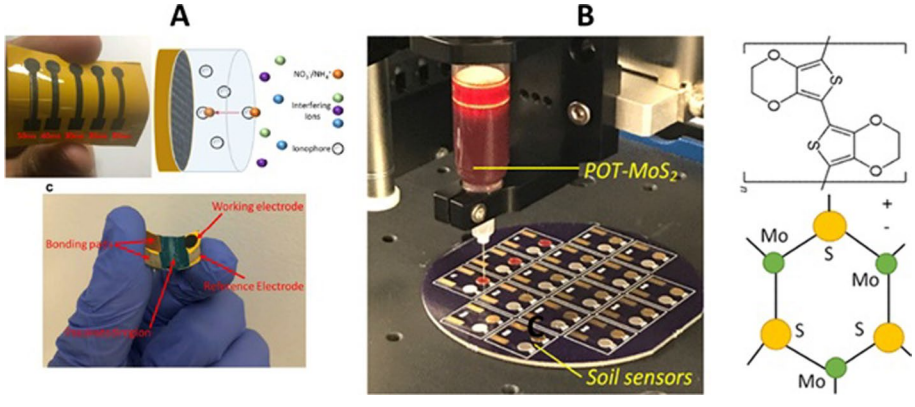

Fig. 5 **a** A laser-induced graphene-based all-solid-state flexible sensor for nitrate and ammonium detection in soil slurries (Copyright Permission ACS 2019). **b** A deployable solid-state sensor on PCB substrate for nitrate-nitrogen detection in soil using a nanocomposite of POT-MoS2 (Copyright Permission ACS 2019)

4 Future needs and prospects

The accurate and temporally resolved mapping of soil properties is the foundational element of precision agriculture, enabling optimized resource use, enhanced economic efficiency, and minimized environmental degradation [1, 78, 148]. While the technological evolution from basic electrical resistance blocks to sophisticated nanosensors and biosensors is well-documented (as detailed in Sects. 2 and 3), the trajectory from promising prototype to ubiquitous, robust field application is fraught with unsolved challenges. This section critically examines the primary technological, scalability, and adoption barriers directly addressing RQ3. We synthesize the current state of development, identify persistent gaps, and propose a roadmap for future research focused on creating integrated, intelligent, and economically viable soil monitoring systems.

4.1 The multi-scale sensing imperative: data integration across spatial domains

A singular technological solution is insufficient to capture the complex heterogeneity of soil systems. The future lies in a synergistic, multi-scale paradigm that strategically integrates data from in-situ, proximal, and remote platforms to create a nested, holistic view of field conditions [189].

In-situ sensors provide the critical foundation of high-temporal-resolution, verifiable ground truth data. Their scalability limitation is not merely numerical but logistical. Deploying a statistically significant number of sensors across a large field is cost-prohibitive and operationally complex. The research focus must, therefore, shift from mere data collection to achieving long-term autonomy and stability. This encompasses innovations in energy harvesting, such as miniaturized solar cells, bio-energy cells utilizing rhizodeposits and microbial activity [22], and piezoelectric generators that harness energy from low-frequency soil vibrations [60]. Concurrently, overcoming signal drift and fouling requires a new generation of self-calibrating sensors. This could involve dual-sensor systems with one active and one reference electrode, or the integration of machine learning (ML) algorithms that continuously correct measurements based on diurnal patterns and cross-correlation with other stable parameters [114]. Proximal sensing platforms, primarily Unmanned Aerial Vehicles (UAVs), serve as the essential intermediary, translating point data into field-scale understanding. The scalability challenge here is not just the unit cost but the data processing bottleneck. The future prospect involves moving beyond raw data collection to intelligent, onboard analysis. Edge computing on UAVs, where hyperspectral and thermal data are processed in real-time to immediately generate prescription maps (e.g., for nitrogen or water deficit), will drastically reduce data latency and transmission costs [167]. Furthermore, sensor fusion integrating optical data with LiDAR for canopy structure and Ground-Penetrating Radar (GPR) for sub-surface moisture mapping will provide a more comprehensive diagnostic picture than any single technology can offer [148].

Satellite-based remote sensing provides the macro-scale context, offering historical trends and monitoring across vast geographies. The primary scalability challenge is the inverse relationship between spatial and temporal resolution. The proliferation of public (Sentinel) and private (Planet Labs) constellations is mitigating this. The future lies in sophisticated AI-driven data fusion techniques. Deep learning models, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), are being trained to synergize data from multiple satellites (e.g., Sentinel-1's SAR with

Sentinel-2's optical data) and to algorithmically remove the confounding effects of vegetation canopy, atmospheric vapor, and surface roughness to isolate the soil signal [155, 192, 194]. The ultimate expression of this multi-scale paradigm is a closed-loop, 'triggered-scouting' system. Satellite data identifies a potential stress zone, an autonomous UAV is dispatched for high-resolution confirmation, and the coordinates are then sent to a robotic ground vehicle equipped with verifiable in-situ sensors for final diagnosis and even targeted intervention (e.g., micro-dosing of fertilizer). This hierarchy of sensing ensures efficiency and maximizes the value of each data point. Table 5 synthesizes the multi-scale soil sensing technologies, their capabilities, scalability challenges, and future innovations.

4.2 Overcoming core hardware challenges: materials, fouling, and energy

The physical interface between the sensor and the soil remains a primary point of failure, directly impacting longevity, accuracy, and cost-effectiveness.

4.2.1 Sensor fouling and drift

Electrochemical sensors (ISEs, ISFETs) are particularly vulnerable to passivation and signal drift caused by the adhesion of soil colloids, organic matter, and biofilms. While Nafion coatings are a step forward, future research must explore more robust anti-fouling strategies. These include biomimetic surfaces inspired by the anti-adhesive properties of shark skin or lotus leaves [53], and the integration of ultrasonic or micro-fluidic systems that create a protective barrier or periodically clean the sensor surface in situ [3]. For drift, the move towards all-solid-state electrodes is critical. The use of novel transducer materials like 3D graphene networks, carbon nanotube matrices, and stable conducting polymers (e.g., PEDOT:PSS) eliminates the internal aqueous solution, providing a more stable reference potential and drastically reducing maintenance needs [101, 213, 214].

4.2.2 Biodegradability and environmental impact

The vision of deploying dense sensor networks raises legitimate concerns about electronic waste (e-waste). A revolutionary approach is the development of transient electronics sensors constructed from biodegradable and eco-friendly materials such as cellulose nanopapers, silk fibroin, and organic semiconductors. These sensors would be designed for a defined operational lifespan (e.g., one growing season) before harmlessly decomposing into the soil, eliminating retrieval costs and environmental impact [89]. This is ideal for single-season monitoring campaigns or distributed sensing at high densities.

4.2.3 Energy autonomy

The power requirement for wireless sensor networks is a fundamental constraint on their scalability. Beyond solar, research into alternative energy harvesting mechanisms is essential. Triboelectric nanogenerators (TENGs) can generate useful power from wind movement across the field or from the impact of raindrops. Microbial fuel cells (MFCs) that generate electricity from the metabolic activity of rhizosphere microbes represent a promising, continuous power source that is intrinsically linked to the soil environment itself [22].

Table 5 Multi-scale soil sensing technologies: capabilities, scalability challenges, and future innovations

Technology tier	Example technologies	Key scalability advantages	Critical scalability challenges (RQ3)	Emerging innovations to address challenges	Key references
In-situ (point)	TDR/FDR, ISFETs, solid-state ISEs, polymer sensors	Gold-standard accuracy for specific points; continuous temporal data; direct measurement of a wide range of parameters (SMC, NPK, pH)	High spatial density required for field representation; labour-intensive installation/maintenance; power supply limitations; signal drift and soil-specific calibration	Energy harvesting (solar, microbial fuel cells); ML-driven self-calibration; ultra-low-power LoRaWAN modules; biodegradable substrates	[22, 60, 114]
Proximal (field)	UAV-mounted hyperspectral, thermal, GPR; EMI sensors	Rapid coverage of entire fields (ha-scale); high spatial resolution (cm-dm); flexible, on-demand deployment; sensor fusion capability	High initial capital cost; regulatory flight restrictions; requires technical expertise; limited soil penetration depth for optical methods; data processing load	Onboard edge computing for real-time analytics; automated flight planning; miniaturized and multi-sensor pods; advanced inversion algorithms for GPR/EMI	[57, 148, 167]
Remote (regional)	Satellite-based SAR (Sentinel-1), Optical/Hyperspectral (Sentinel-2, PRISMA)	Unmatched spatial extent; long-term data archives; free data access (Sentinel); ideal for trend analysis and large-area monitoring	Coarse spatial resolution (10 m-1 km); infrequent revisit time; signals attenuated by vegetation and atmosphere; surface-only measurement	AI/ML for data fusion and vegetation correction; new constellations with higher resolution; integration with climate models	[28, 155, 192, 194]
Robotic/automated	Autonomous ground vehicles (AGVs), robotic soil samplers	Automated high-resolution mapping; can carry heavy sensor payloads; potential for direct intervention (e.g., micro-dosing)	Very high development cost; navigation challenges in complex, wet terrain; limited battery life and field endurance	Swarm robotics for distributed tasks; advanced computer vision and LiDAR for navigation; autonomous docking stations for charging	[123, 165]
Next-gen bio/chem	Nanosensors, CRISPR-based biosensors, synthetic biology reporters	Potential for ultra-high sensitivity and specificity; detection of complex biological properties (e.g., specific microbial functions, enzyme activity)	Predominantly in R&D phase; unknown long-term stability and drift in complex soil matrices; potential ecological risk; complex readout requirements	Development of single-use, disposable designs; integration with smartphone-based readouts; extensive field validation studies	[49, 133, 161]

4.3 From data to wisdom: analytics, interoperability, and decision support

The collection of data is meaningless without its transformation into timely, actionable intelligence. This represents perhaps the most significant software and scalability challenge.

4.3.1 Interoperability and open standards

The current landscape is fragmented, with proprietary data formats and closed communication protocols from different manufacturers creating data silos. Widespread adoption hinges on the embrace of open-source standards. This includes communication protocols like MQTT for IoT messaging, data ontologies like AGROVOC for universal agricultural terminology, and platforms like FIWARE for managing context data [95, 96]. This allows a sensor from brand A to seamlessly integrate with a gateway from brand B and an analytics platform from brand C.

The Edge-Cloud Continuum Transmitting raw, high-frequency data from thousands of sensors to the cloud is impractical. The future lies in a hybrid edge-cloud computing architecture. Edge computing modules located on the UAV or field gateway perform initial data processing: filtering noise, extracting features, and identifying significant events (e.g., a sudden drop in moisture). This reduces the data volume by orders of magnitude, sending only summary statistics or alerts to the cloud. The cloud platform then performs heavier-duty tasks: long-term data storage, integration with historical weather and yield data, and running complex process-based models for predictive analytics [108].

Digital Twins and Prescriptive Analytics The next evolution is moving from descriptive ("what is happening") and predictive ("what will happen") analytics to prescriptive analytics ("what should I do"). This is enabled by creating a digital twin of the field a dynamic, virtual model that is continuously updated with real-time sensor data. Farmers and agronomists can use this twin to run simulations, testing the outcome of different management decisions (e.g., "if I irrigate with 20 mm tomorrow, what will the soil moisture be next week?") before committing resources in the real world. This de-risks decision-making and optimizes outcomes [113].

4.4 Economic and human-centric adoption barriers

Technology that is not adopted is irrelevant. Scaling requires overcoming cost barriers and ensuring user trust.

4.4.1 Novel business models

The high capital expenditure (CAPEX) is a primary barrier, especially for smallholders. Sensor-as-a-Service (SeaaS) models, where farmers pay a monthly or annual subscription fee for data insights without owning the hardware, can lower the initial barrier to entry [92]. The service provider owns, maintains, and calibrates the sensor network, guaranteeing data quality. Furthermore, better quantification of Return on Investment (ROI) is needed, clearly demonstrating savings in water, fertilizer, and pesticides, alongside yield increases and potential premium prices for sustainably certified produce.

4.4.2 User-centric design and digital literacy

Interfaces must be designed for farmers, not engineers. Dashboards should be visual, intuitive, and provide clear, actionable recommendations (e.g., 'Apply 50 kg/ha of K_2O in the red zone') rather than raw data streams. Concurrently, significant investment in digital extension services is required to build capacity, trust, and digital literacy within the farming community, ensuring they can effectively interpret and act upon the data provided [21].

Table 6 Quantitative comparisons of metrological parameters [30, 128, 200]

Sensor type	Accuracy	Precision	Sensitivity	Resolution	Uncertainty	Response time	Drift
TDR	High (± 3%)	0.005–0.02	High	0.001	Low	Fast (1–60 s)	Low
FDR	Moderate (± 3–6%)	0.01–0.03	Medium	0.002	Medium	Fast (1–30 s)	Medium
EC sensors	Moderate	0.02–0.05	High	0.01	Medium	1–10 s	Low
ISE (nitrate)	Moderate	0.02–0.1	High	0.01	High	30–300s	High
Optical VIS–NIR	Moderate–High	0.01–0.05	Variable	0.005	Medium	1–10 s	Medium
MEMS/biosensor	Variable	0.01–0.05	High	0.001	Medium	1–10 s	High

4.4.3 Policy and ecosystem incentivization

Governments and international bodies can act as powerful accelerators. Policies could include subsidies for adopting precision agriculture practices verified by sensor data, carbon credit programs that rely on sensor-generated data for accurate quantification of soil carbon sequestration, and the development of open-access sensor networks as public goods for regional monitoring [202].

4.5 Quantitative collation of metrological parameters in soil sensing technology

Metrological parameters can vary in their performance due to the variety of measurement methods, which include electrical, optical, electrochemical, and mechanical. Sensor suitability for various field and laboratory applications is assessed by quantitative comparisons of metrological parameters. The reliability with which a sensor measures is determined by metrological characterisation. When we need to assess and compare the performance of soil sensors like TDR, FDR, EC, ion-selective, and optical sensors, factors like accuracy, precision, resolution, sensitivity, reaction time, uncertainty, and drift are critical [149]. By using adaptive uncertainty estimate, AI integration improves metrological traceability. According to theoretical models, AI can minimise effective measurement uncertainty (Ueff) in dynamic settings by 15–40% [5]. Beyond conventional single-sensor limitations, sensor fusion allows multi-parameter estimate (such as moisture-salinity-temperature coupling). AI-driven correction lowers calibration frequency and accounts for more soil texture heterogeneity, according to theoretical studies. It is essential to provide a comparative metrological basis for various types of soil sensors and to emphasise AI's revolutionary role in sensor calibration, uncertainty reduction, and intelligent metrological traceability [38]. In order to assure data accuracy, precision, and dependability across a variety of soil settings, future soil sensing systems may have to depend more and more on hybrid AI–metrology models (Table 6).

4.6 Emerging frontiers: GHG monitoring and soil biology

The future of soil sensing extends beyond traditional agronomy into environmental accountability and advanced biology.

4.6.1 Greenhouse Gas (GHG) flux monitoring

There is a pressing need to accurately quantify agriculture's contribution to climate change through nitrous oxide (N_2O) and methane (CH_4) emissions. The development

of low-cost, field-deployable photoacoustic spectroscopy sensors and electrochemical microsensors for continuous, in-situ GHG monitoring is a critical research frontier. This will allow for precise cause and effect monitoring of how management practices (e.g., fertilizer timing, tillage) directly influence GHG fluxes, enabling true climate-smart agriculture [1].

4.6.2 Soil biological activity sensing

Current sensors largely ignore the biological component of soil health the most dynamic and complex fraction. Future prospects include enzyme-specific biosensors that detect the activity of key enzymes like phosphatase or β-glucosidase, providing a direct measure of nutrient mineralization rates. CRISPR-based biosensors that can detect the DNA of specific beneficial or pathogenic microorganisms directly in the soil. CRISPR-based biosensors operate on the principle of a naturally occurring defensive immune mechanism in prokaryotes, which involves CRISPR arrays and Cas proteins [160]. They are known for detecting nucleic acid targets with high sensitivity and specificity [77, 211]. Emerging technologies have enhanced these sensors, as they can be integrated with microfluidic devices and isothermal amplification methods to detect targets at even low concentrations [201]. Respiration sensors that measure real-time microbial CO_2 flux as a proxy for overall biological activity and organic matter decomposition [133, 161].

4.6.2.1 Role of Drone technology in soil sensing To eradicate the devastating crop losses due to pests and diseases, early detection of microbial diseases would be the most important strategy to be implemented [23]. In such scenarios, drones equipped with sensors come into the picture as a boon. It can monitor crop health by scanning the crops using green visible and near-IR light, and through real-time monitoring, it can help in optimising the use of inputs and resources, hence, preventing ecological degradation [171]. According to FAO, "Drones equipped with specific sensors can collect multispectral images that are stitched to generate spectral reflectance bands. These bands allow users to calculate indices such as a Normalized Difference Vegetation Index (NDVI), a Leaf Area Index (LAI) or a Photochemical Reflectance Index (PRI), allowing farmers to view crop changes or stress conditions that are otherwise invisible to the human eye" [179].

This would move soil health assessment from a purely physico-chemical paradigm to a truly holistic, ecological one. In addressing RQ3 requires a paradigm shift in perspective. The challenge is no longer merely a technological one of developing a more sensitive or cheaper sensor. It is a systems integration challenge that spans materials science, data analytics, energy engineering, human–computer interaction, and economics. The path to scalability is paved with robustness (creating sensors that are resilient to the harsh soil environment), intelligence (embedding processing power to transform raw data into actionable insights at the source), interoperability (adopting open standards to create a cohesive ecosystem, not a collection of isolated gadgets), and accessibility (developing new business models and user interfaces that make the technology economically and practically viable for all farmers. The next decade of research must be intensely interdisciplinary, focused not on isolated breakthroughs but on the seamless integration of hardware, software, and human factors. The goal is to deliver not just data, but wisdom empowering farmers to steward their soil resources towards a more productive, sustainable, and resilient agricultural future.

5 Need for standardization and calibration in soil sensor systems

The growing diversity of soil sensing technologies-varying in sensing principles, data structures, metadata conventions, and calibration methodologies has introduced substantial fragmentation that hinders interoperability across devices, platforms, and analytical systems. While the increasing availability of sensors capable of measuring soil moisture, electrical conductivity, temperature, pH, nutrient status, and dielectric properties has expanded monitoring capabilities, the lack of unified data formats, consistent communication layers, and standardized calibration procedures restricts their integration into decision-support tools and AI-driven precision agriculture frameworks. These limitations have emerged as a central bottleneck for practical deployment. As noted by Chung et al. [40], progress toward interoperable soil sensor ecosystems requires coordinated standardization efforts that span data representation, communication infrastructure, and calibration traceability, engaging sensor manufacturers, researchers, and international bodies such as ISO, OGC, and IEEE.

5.1 Standardization of data formats and schemas

The absence of harmonized data formats and schemas remains one of the most fundamental impediments to interoperability. Soil sensors currently produce outputs that range from analog voltage signals to structured digital datasets containing spatial coordinates, timestamps, calibration parameters, and environmental metadata. Such diversity results in inconsistent variable definitions, differing units, varying temporal resolutions, and incomplete metadata, which complicate cross-sensor comparison and restrict seamless data fusion. Standards such as ISO 28258, which provides XML-based structures for soil data exchange, and OGC frameworks including SensorThings API, SensorML, and Observations & Measurements (O&M), offer robust pathways for representing sensor observations, device metadata, and calibration history in a consistent manner. The INSPIRE Directive similarly provides harmonized data models for environmental datasets, facilitating interoperable soil and land information systems [25, 162]. Despite these advances, commercial soil sensors rarely adopt such standards, leading to semantic inconsistencies between optical, electrochemical, and dielectric systems and causing metadata loss when integrating multi-sensor datasets. Many devices fail to document calibration information, uncertainty bounds, soil physical parameters, or raw signal outputs necessary for reproducibility. As noted by Barbedo [24], adopting linked data and harmonized semantic frameworks is increasingly necessary to ensure cross-domain integration of soil, crop, and climate data, particularly for AI-driven analytics that rely on consistent metadata structures.

5.2 Standardization of communication protocols and messaging layers

Interoperability is further undermined by inconsistencies in communication and messaging protocols used across soil sensing platforms. Current deployments use a mixture of low-power long-range systems such as LoRaWAN and NB-IoT, mesh-based networks such as Zigbee, and internet-layer protocols including HTTP, MQTT, and CoAP. Although each protocol addresses specific operational needs, their uncoordinated implementation results in incompatible message structures, inconsistent encoding of sensor observations, and variable quality-of-service characteristics. For soil sensors integrated into IoT-based agricultural systems, uniformity in communication layers is

essential to ensure reliable, secure, and temporally consistent data exchange. Messaging standards must ensure that sensor observations, metadata, calibration updates, and timestamps are encoded and transmitted in a coherent manner across cloud, edge, and device layers. Equally central is measurement traceability, by which sensor outputs must be reliably linked to SI-referenced standards or acknowledged benchmark methods to ensure cross-platform comparability [39]. Without such traceability, multi-site comparisons and long-term environmental monitoring lose scientific integrity. Standardized communication layers also support the implementation of advanced calibration solutions, including automated drift correction or metadata-based recalibration workflows, which rely on consistent real-time data transmission.

5.3 Standardization of calibration and traceability procedures

Calibration remains the most technically critical yet least standardized aspect of soil sensor systems. Soil heterogeneity arising from variations in texture, bulk density, salinity, temperature, and organic matter significantly influences sensor response, making universal calibration curves inadequate for most sensor types. Robust calibration generally proceeds through several scientific stages, including the mapping of raw sensor outputs to intermediate physical quantities such as dielectric permittivity, transforming these quantities into agronomic parameters such as volumetric water content, adjusting for confounding environmental factors, and validating the resulting outputs against reference methods such as gravimetric analysis or TDR [111]. Laboratory calibrations, though controlled, often diverge from field behaviour due to spatial heterogeneity, variable moisture profiles, and dynamic temperature conditions. Consequently, field validation and periodic recalibration remain indispensable. Ageing, fouling, and environmental exposure introduce sensor drift, and reproducibility in the literature is often compromised by inadequate reporting of calibration metadata, raw signals, or environmental context [210].

To overcome these limitations, soil sensor systems must provide transparent access to raw outputs, document calibration coefficients and environmental conditions in machine-readable formats, and apply consistent cross-sensor calibration using certified reference instruments [151]. Factory calibration, cross-sensor field calibration, and in-situ recalibration under changing soil conditions constitute the core components of traceable calibration workflows [116]. Increasingly, real-time drift correction is enabled through AI-based auto-calibration models that employ machine learning and transfer-function mapping to harmonize sensor outputs across heterogeneous devices [109, 172]. Such algorithmic calibration approaches further highlight the need for standardized data formats and metadata structures, as AI models cannot reliably correct sensor drift or harmonize outputs without consistently structured calibration data. Therefore, genuine interoperability in soil sensing requires integrated standardization across data formats, communication protocols, and calibration procedures. Without such standardization, sensor networks will remain fragmented, limiting their scientific and practical utility. Establishing common schemas, harmonized communication frameworks, and transparent, traceable calibration standards will be essential for enabling large-scale soil monitoring, multi-sensor data fusion, and the development of globally consistent soil intelligence infrastructures.

6 Conclusion

In this comprehensive review, we have critically examined the transformative role of advanced sensor system technologies in soil health monitoring, addressing the three core research questions that guided the inquiry. The synthesis of recent advancements reveals a paradigm shift from rudimentary, lab-dependent methods toward sophisticated, integrated systems capable of delivering real-time, high-resolution data essential for precision agriculture. In response to RQ1, the review delineates how breakthroughs in material science and transducer design have fundamentally enhanced sensor capabilities. Innovations in microelectromechanical systems (MEMS), nanotechnology, and polymer-based composites have yielded sensors with markedly improved accuracy, specificity, and field-deployability for critical parameters such as moisture, NPK nutrients, pH, and temperature. The evolution from conventional glass electrodes to solid-state Ion-Selective Field-Effect Transistors (ISFETs) and optical sensors exemplifies this progress, offering greater durability and precision for in-situ measurements. Similarly, dielectric techniques like TDR and FDR have been refined to provide reliable soil moisture data with reduced susceptibility to environmental interference. Addressing RQ2, the analysis confirms that integrated platforms, notably Wireless Sensor Networks (WSNs) and the Internet of Things (IoT), are effectively overcoming the spatial and temporal limitations inherent in traditional point-based sensing. These platforms facilitate extensive, continuous data collection and enable predictive analytics through cloud computing and machine learning integration. The synergy between proximal sensing (e.g., UAV-mounted hyperspectral imaging) and remote sensing technologies provides a multi-scale understanding of field variability, supporting data-driven decision-making for irrigation and nutrient management on a previously unattainable scale. However, the investigation into RQ3 underscores that the path from research prototype to widespread adoption is fraught with significant challenges. Key technological barriers persist, including the imperative for soil-specific calibration to ensure accuracy across diverse pedological conditions, the high initial costs of advanced sensor systems, and the limited connectivity in remote agricultural regions. Scalability is further hampered by issues of sensor fouling, signal drift, and energy autonomy for wireless networks. The transition to robust, cost-effective tools necessitates future research focused on developing self-calibrating sensors, universal calibration models leveraging machine learning, and innovative energy harvesting solutions such as solar cells and microbial fuel cells. Furthermore, the development of biodegradable sensor materials could mitigate environmental concerns associated with large-scale deployment. The future of soil sensing lies in a holistic, multi-scale paradigm that strategically combines in-situ, proximal, and remote sensing data. This integration, supported by advances in AI-driven data fusion and prescriptive analytics, will be crucial for creating closed-loop systems for precision agriculture. Ultimately, while sensor technologies hold immense potential for revolutionizing sustainable soil management, realizing this potential requires concerted interdisciplinary efforts to address the existing technological, economic, and adoption barriers, thereby bridging the critical gap between innovative prototyping and practical, large-scale implementation.

Authors contribution

U. T.: Conceptualization, Literature search and review, Writing- original draft; D. P.: Conceptualization, Literature search and review, Writing- section; A. G. B.: Literature search and review, Writing- section; V. T.: Literature search and review, Writing- section; D. N. P.: Literature search and review, Writing- section; S. D.: Literature search and review, Writing- section;

Editing, S. G.: Conceptualization, Literature search and review, Writing- original draft, Editing and formatting, Supervision, Resources. All co-authors read, commented, and approved the final manuscript.

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